

A Symbolic–Affine Framework for Density-One Collatz Descent

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October 8, 2026

Abstract

We study finite stopping time for the Collatz map through the fully accelerated transformation

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

acting on the odd positive integers.

Finite accelerated trajectories are encoded by *division words*, which record the successive 2-adic valuations encountered along an orbit. Each division word determines a formal affine map

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}.$$

For every nonempty word with positive entries, exact realization is equivalent to a single affine congruence modulo $2^{D(\omega)+1}$. Consequently, its realizations form a unique dyadic cylinder of natural density $2^{-(D(\omega)+1)}$, and symbolic prefix extension corresponds exactly to cylinder containment.

We organize these cylinders into a residual forest according to the first prefix at which the affine slope becomes strictly less than 1. Residual words have affine slope strictly greater than 1 at every nonempty prefix, whereas first-contraction words acquire subunit affine slope for the first time at their terminal position. The resulting residual sets satisfy an exact finite-stage decomposition into first-contraction layers and continued residual survival.

To prove that the residual mass vanishes, we introduce the auxiliary weight

$$\left(\frac{3}{8}\right)^{D(\omega)},$$

whose total residual weight contracts by a factor at most $3/5$ at each symbolic step. Comparison with the true dyadic cylinder weights, together with

$$B(m+5) \leq B(m) + 8,$$

yields a five-step envelope contraction with factor

$$\left(\frac{4}{3}\right)^8 \left(\frac{3}{5}\right)^5 = \frac{65536}{84375} < 1.$$

Hence the natural densities of the finite-depth residual sets tend to zero.

A uniform cutoff on each first-contraction layer converts affine contraction into genuine descent apart from finitely many exceptional values.

Since the initial residual class is exactly $3 \bmod 4$, while positive even integers descend after one ordinary Collatz step and odd integers congruent to $1 \bmod 4$, apart from 1, descend after one fully accelerated step, it follows that the set of positive integers having an ordinary Collatz iterate below their starting value has natural density one.

The proof is deterministic, uses finite-stage natural-density accounting, and is independently formalized in Lean 4.

Contents

1	Introduction	5
2	Conventions	10
3	Division Counts and Division Words	11
3.1	Division Counts and Division Words	11
3.2	Word Length and Total Division Count	13
4	Affine Representation of Division Words	14
4.1	Formal Symbolic Action and Affine Representation	14
4.2	Concatenation and Prefix Structure	17
4.3	Explicit Form of the Affine Constant	19
4.4	Dyadic-Ternary Balance	22
4.5	Formal Integrality versus Exact Realization	25
5	Exact Realizability and Residue Structure	28
5.1	Affine Integrality and Prefix Propagation	28
5.2	Exact Valuation and the Additional Parity Bit	31
5.3	The Exact Realization Criterion	32
5.4	Exact Realization Residue Classes	35
5.5	Nested Structure of Realization Classes	37
6	Cylinder Geometry of Division Words	40
6.1	Dyadic Cylinders and Division-Word Cylinders	40
6.2	Geometry of Dyadic Cylinders	41
6.3	Counting and Natural Density	43
6.4	Realization Cylinder Forest	45
7	Affine Contraction and Descent Cylinders	48
7.1	Affine Contraction and Pointwise Descent	49
7.2	Asymptotic Descent Cylinders	51
8	First-Contraction and the Residual Forest	53
8.1	First-Contraction and Residual Words	54
8.2	Residual Levels and Exit Layers	55
8.3	Residual Mass and Finite-Stage Conservation	60
9	Residual Mass Collapse	62

9.1	Auxiliary Weighted Residual Mass	63
9.2	Transfer to the Dyadic Residual Mass	65
9.3	Five-Step Envelope Contraction	67
9.4	Residual Mass Collapse and First-Contraction Exhaustion	69
10	Density-One Eventual Descent	71
10.1	Uniform Descent on First-Contraction Layers	71
10.2	Density Zero of Residual Non-Descent	74
10.3	The Initial Residual Congruence Class	76
10.4	Transfer to the Ordinary Collatz Map	78
10.5	Density-One Descent	79
11	Conclusion	82
A	Explicit Computation of the Affine Map	85
A.1	A Four-Step Example	86
A.2	Direct Derivation by Symbolic Substitution	87

1 Introduction

The $3x + 1$ problem, commonly known as the Collatz conjecture, concerns iteration of the map

$$T(x) = \begin{cases} \frac{x}{2}, & x \text{ even,} \\ 3x + 1, & x \text{ odd,} \end{cases}$$

on the positive integers. The conjecture asserts that every positive integer eventually reaches 1. Despite its elementary formulation, the problem has resisted proof for decades; historical accounts and surveys of the problem and its many equivalent formulations are given by Lagarias [1]. Extensive computation has also verified convergence over large finite ranges [2].

A classical weakening of the conjecture asks only whether an orbit eventually falls below its starting value. The least positive iterate at which this occurs is commonly called the *stopping time*. Terras [3] and, independently, Everett [4] established that finite stopping time occurs for almost every positive integer in the sense of natural density.

Several ideas underlying such finite-stopping arguments are already present in the classical work. Terras encodes finite trajectories by parity data and expresses the resulting iterates in affine form, with a multiplicative coefficient determined by that finite symbolic data together with an additive remainder. The associated *coefficient stopping time* records the first depth at which the multiplicative coefficient becomes less than 1. The finite parity data also determine arithmetic progressions of starting values, providing a residue-class structure from which density statements can be obtained; see [3, 1]. Thus affine trajectory representations, finite symbolic coding, residue-class organization, and the density-one finite-stopping theorem itself are classical ingredients.

Subsequent work has obtained substantially stronger control of orbit minima in other density settings. Tao [5] proved that, in logarithmic density, almost every Collatz orbit attains almost bounded values: for every function $g(N) \rightarrow \infty$, the minimum value attained by the orbit of N is at most $g(N)$ for logarithmic-density almost every N .

The purpose of the present work is therefore not to introduce affine trajectory coding or to claim a new density-one finite-stopping theorem. Rather, it develops a self-contained deterministic formulation of the finite-stopping argument in fully accelerated coordinates and organizes the proof around exact realization classes and a

residual-cylinder decomposition. The odd part of the dynamics is encoded using the fully accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

which maps odd positive integers to odd positive integers by removing all factors of 2 following each $3x+1$ step.

A finite trajectory under f determines a *division word*

$$\omega = (d_0, \dots, d_{m-1}), \quad d_j = v_2(3f^j(x) + 1),$$

recording the exact powers of 2 removed at successive accelerated steps. Write

$$D(\omega) = d_0 + \dots + d_{m-1}$$

for its total division count. Symbolic composition associates to every division word the formal affine map

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}.$$

This is the fully accelerated analogue of the classical affine description of finite Collatz trajectories. It is algebraic and is defined independently of whether the prescribed word is realized by an actual accelerated trajectory.

In these coordinates the arithmetic realization problem has an exact congruential form. For an admissible division word—that is, a nonempty word whose entries are positive—a natural number x realizes ω if and only if

$$3^{|\omega|}x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}.$$

Consequently, the realization set of ω is a unique residue class modulo

$$2^{D(\omega)+1}.$$

In particular, every admissible finite division word is realizable. The additional factor of 2 in the modulus distinguishes exact accelerated realization from the weaker condition that the formal affine expression is merely integral.

These realization classes form nested dyadic cylinders. If C_ω denotes the realization cylinder of ω , then

$$\text{dens}(C_\omega) = 2^{-(D(\omega)+1)},$$

and symbolic prefix extension corresponds exactly to cylinder containment. Prefix-incomparable words determine disjoint cylinders. Thus the classical residue-class

organization of finite trajectories takes, in the present accelerated coordinates, the form of a nested dyadic cylinder geometry.

The affine slope of a division word is

$$\lambda_\omega = \frac{3^{|\omega|}}{2^{D(\omega)}}.$$

Let

$$B(m) = \lfloor m \log_2 3 \rfloor + 1$$

denote the least integer k for which

$$3^m < 2^k.$$

Then

$$\lambda_\omega < 1 \iff D(\omega) \geq B(|\omega|).$$

This is the accelerated-coordinate counterpart of the classical coefficient-stopping comparison: it concerns the multiplicative affine coefficient rather than pointwise descent itself.

A subunit affine slope does not by itself imply descent for every realization, because the positive affine constant may dominate for small initial values. Nevertheless,

$$F_\omega(x) < x$$

for every sufficiently large realization of a fixed subunit-slope word. Hence only finitely many points of such a realization cylinder can fail to lie below their starting value at the endpoint of the encoded trajectory segment, and removing those points does not change the cylinder's natural density.

The global argument is organized by the first prefix at which this affine threshold is reached. An admissible word is called *residual* when every nonempty prefix remains above unit affine slope, and a *first-contraction word* when its proper prefixes are residual but its terminal prefix has subunit slope. Although the full family of admissible words is infinite at every positive symbolic length, the residual family at each fixed depth is finite.

If $\mathcal{R}_m$ denotes the union of the realization cylinders of residual words of length m , and $\mathcal{F}_{m+1}$ the corresponding first-contraction exit layer, then

$$\mathcal{R}_m = \mathcal{F}_{m+1} \sqcup \mathcal{R}_{m+1}.$$

This exact finite-stage decomposition provides the conservation law used throughout the residual-forest argument.

Writing

$$M_m = \text{dens}(\mathcal{R}_m)$$

for the residual mass, the quantitative task is to prove

$$M_m \longrightarrow 0.$$

The proof introduces the auxiliary weight

$$\left(\frac{3}{8}\right)^{D(\omega)},$$

whose total weight satisfies the uniform one-step bound

$$W_{m+1} \leq \frac{3}{5} W_m.$$

Comparison with the true dyadic cylinder weights, together with the elementary threshold estimate

$$B(m+5) \leq B(m) + 8,$$

yields a five-step envelope contraction with factor

$$\left(\frac{4}{3}\right)^8 \left(\frac{3}{5}\right)^5 = \frac{65536}{84375} < 1.$$

It follows that the unresolved residual mass tends to zero.

The final step converts affine first contraction into genuine dynamical descent. Although a fixed first-contraction layer may contain infinitely many terminal words, its residual parent level is finite. The affine concatenation formula therefore supplies a uniform finite cutoff above which every point in the entire exit layer descends. At any fixed residual depth, the set of non-descending points is consequently contained in the remaining residual set together with a finite exceptional set. Taking the residual depth first and then the interval size shows that initially residual non-descent has natural density zero.

The initial residual set itself has the simple description

$$\mathcal{R}_1 = \{x : x \equiv 3 \pmod{4}\}.$$

Positive even integers descend immediately under T , while odd integers congruent to 1 (mod 4), apart from 1, descend after one fully accelerated step. Accelerated iterates occur along the ordinary Collatz orbit, so accelerated descent transfers to ordinary descent. Thus the residual argument handles the remaining nontrivial congruence class and recovers the classical density-one finite-stopping theorem:

$$\text{dens} \{x \in \mathbb{N}_{>0} : \exists n, T^n(x) < x\} = 1.$$

The resulting proof is organized through the sequence

division words $\longrightarrow$ affine realization $\longrightarrow$ dyadic cylinders
 $\longrightarrow$ residual forest $\longrightarrow$ residual mass collapse $\longrightarrow$ eventual descent.

This organization should be viewed as a fully accelerated, exact-arithmetic reformulation of classical finite-stopping machinery rather than as a claim of priority for affine or residue coding. Its distinguishing role in the present proof is to isolate the unresolved prefixes into finite residual levels and control their total dyadic mass through the residual-forest decomposition. No probabilistic model of trajectory independence is used, and the density argument proceeds through finite-stage decompositions and ordinary natural-density limits.

The result concerns descent below the starting value, not convergence to 1. It does not assert that every positive starting value has finite stopping time, nor does it prove that every Collatz trajectory converges. The density-zero complement of the descent set may still be nonempty or infinite.

Formal verification and reproducibility. The Lean 4 formalization accompanying this manuscript is available at

<https://github.com/wbrassem/collatz-density-one-descent>

The culminating formal theorem is

`Collatz.ordinary_collatz_descent_has_natural_density_one`

The repository contains the Lean project, verification harness, continuous-integration workflow, and manuscript source.

Structure of the Paper. The argument proceeds as follows:

- Section 3 introduces division words, realization, and total division counts.

- Section 4 develops the formal affine representation and the dyadic–ternary slope threshold.
- Section 5 proves the exact realization criterion and the unique residue class associated with each admissible word.
- Section 6 develops generic dyadic-cylinder geometry, counting, natural density, and the realization-cylinder forest.
- Section 7 relates subunit affine slope to pointwise descent and proves density preservation after removing the finite affine exceptional set.
- Section 8 introduces residual and first-contraction words and establishes the finite-stage forest and mass decompositions.
- Section 9 proves that the residual mass tends to zero by means of the auxiliary weighted mass and the five-step envelope contraction.
- Section 10 converts first contraction into genuine descent, identifies the initial residual congruence class, and transfers the accelerated result to the ordinary Collatz map.

2 Conventions

Throughout,

$$\mathbb{N} = \{0, 1, 2, \dots\}, \quad \mathbb{N}_{>0} = \{1, 2, 3, \dots\}.$$

The classical Collatz conjecture concerns iteration on $\mathbb{N}_{>0}$. The ordinary Collatz map is therefore regarded as a dynamical map on positive integers, while the fully accelerated Collatz map acts dynamically on the odd positive integers.

For the arithmetic and symbolic constructions used in this paper, however, it is convenient to work in the ambient set $\mathbb{N}$. Generic residue classes, dyadic cylinders, symbolic data, exponents, indices, and counting functions are therefore allowed to involve 0.

Natural density is computed using the initial intervals

$$[0, N) = \{0, 1, \dots, N - 1\}.$$

Thus a subset of $\mathbb{N}_{>0}$ is also regarded as a subset of $\mathbb{N}$ for purposes of density; the omission of the single point 0 has no effect on its natural density.

Division words may contain the value 0. Positivity conditions required for accelerated odd trajectories are imposed explicitly through admissibility rather than through the ambient definition of $\mathbb{N}$.

Thus the occurrence of 0 in the symbolic and arithmetic constructions does not enlarge the dynamical domain of the Collatz problem.

3 Division Counts and Division Words

3.1 Division Counts and Division Words

For a positive integer n , let $v_2(n)$ denote the largest $k \geq 0$ such that $2^k \mid n$. For $x \in \mathbb{N}$, define the *division count*

$$d(x) := v_2(3x + 1),$$

and define

$$f(x) := \frac{3x + 1}{2^{d(x)}}.$$

These formulas define arithmetic functions on all of $\mathbb{N}$. Restricted to the odd positive integers, f is the fully accelerated Collatz map.

Since $2^{d(x)} \mid 3x + 1$, the value $f(x)$ is again a natural number. By construction, removing the complete power of 2 from $3x + 1$ leaves an odd integer. Hence

$$f(x) \text{ is odd for every } x \in \mathbb{N}.$$

Moreover, if x is odd, then $3x + 1$ is even, and therefore

$$d(x) \geq 1.$$

Definition 3.1 (Division Word). A *division word* is a finite sequence

$$\omega = (d_0, \dots, d_{m-1}),$$

where $m \geq 0$ and each $d_i \in \mathbb{N}$. The case $m = 0$ is the empty division word, denoted by $\emptyset$.

For $x \in \mathbb{N}$ and $m \geq 0$, define the length- m division word generated by x to be

$$\omega_m(x) := (d(x), d(f(x)), \dots, d(f^{m-1}(x))),$$

with

$$\omega_0(x) = \emptyset.$$

Definition 3.2 (Valid Division Word). A division word

$$\omega = (d_0, \dots, d_{m-1})$$

is *valid* if it is nonempty, equivalently if

$$m \geq 1.$$

Definition 3.3 (Admissible Division Word). A division word

$$\omega = (d_0, \dots, d_{m-1})$$

is *admissible* if it is valid and

$$d_i \geq 1, \quad 0 \leq i < m.$$

Definition 3.4 (Realization). Let

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word. A natural number $x \in \mathbb{N}$ *realizes* ω if

$$\omega = \omega_m(x).$$

Equivalently,

$$d_i = v_2(3f^i(x) + 1), \quad 0 \leq i < m.$$

A division word is called *realizable* if it is realized by at least one natural number.

Lemma 3.1 (Generated Division Words). *For every $x \in \mathbb{N}$ and every $m \geq 0$, the generated word $\omega_m(x)$ is a division word of length m and is realized by x .*

If x is odd and $m \geq 1$, then $\omega_m(x)$ is admissible.

Proof. For every $i \geq 0$,

$$d(f^i(x)) \in \mathbb{N},$$

so $\omega_m(x)$ is a division word. Its length is m , and realization by x follows directly from its definition.

Now suppose that x is odd and $m \geq 1$. Then

$$d(x) \geq 1.$$

Every subsequent iterate $f^i(x)$ is odd, because f removes the complete power of 2 from the corresponding numerator. Hence

$$d(f^i(x)) \geq 1, \quad 0 \leq i < m.$$

Thus every entry of the nonempty word $\omega_m(x)$ is positive, and $\omega_m(x)$ is admissible. $\square$

Remark. Realization and admissibility are distinct notions. Admissibility is an intrinsic positivity condition on a nonempty symbolic word, whereas realization asserts that the word actually occurs as an initial segment generated by iteration of f .

In particular, no converse to the preceding lemma is being assumed here. The arithmetic characterization of realizable division words will be established later.

3.2 Word Length and Total Division Count

Definition 3.5 (Length). Let

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word. Its *length* is

$$|\omega| := m.$$

Definition 3.6 (Total Division Count). Let

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word. Its *total division count* is

$$D(\omega) := \sum_{i=0}^{m-1} d_i.$$

For the empty word,

$$D(\emptyset) = 0.$$

Lemma 3.2 (Length Bound). *Every admissible division word satisfies*

$$|\omega| \leq D(\omega).$$

Proof. If

$$\omega = (d_0, \dots, d_{m-1})$$

is admissible, then $d_i \geq 1$ for every $0 \leq i < m$. Hence

$$D(\omega) = \sum_{i=0}^{m-1} d_i \geq \sum_{i=0}^{m-1} 1 = m = |\omega|.$$

□

Remark. The quantities

$$|\omega| \quad \text{and} \quad D(\omega)$$

record different aspects of the symbolic trajectory. The length $|\omega|$ is the number of applications of f encoded by the word, whereas $D(\omega)$ is the cumulative exponent of 2 removed across those steps.

Their interaction will determine the linear coefficient

$$\frac{3^{|\omega|}}{2^{D(\omega)}}$$

of the affine representation developed below.

4 Affine Representation of Division Words

4.1 Formal Symbolic Action and Affine Representation

Definition 4.1 (Symbolic Step). For a prescribed division count $d \in \mathbb{N}$, define the *symbolic step*

$$S_d : \mathbb{Q} \rightarrow \mathbb{Q}, \quad S_d(x) := \frac{3x + 1}{2^d}.$$

No divisibility or realizability condition is imposed. Thus S_d is defined for every $d \in \mathbb{N}$ and every $x \in \mathbb{Q}$.

Definition 4.2 (Symbolic Action of a Division Word). Let

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word. Its *symbolic action* is the composition

$$S_\omega := S_{d_{m-1}} \circ \dots \circ S_{d_0}.$$

For the empty word,

$$S_{\emptyset}(x) := x.$$

Equivalently, if

$$\omega = (d_0, \eta),$$

where η denotes the remaining tail, then

$$S_{\omega}(x) = S_{\eta}(S_{d_0}(x)).$$

Definition 4.3 (Affine Constant). The affine constant $c(\omega) \in \mathbb{N}$ of a division word is defined recursively by

$$c(\emptyset) = 0,$$

and, for

$$\omega = (d_0, \dots, d_{m-1}), \quad m \geq 1,$$

by

$$c(\omega) = 3^{m-1} + 2^{d_0}c(d_1, \dots, d_{m-1}).$$

Definition 4.4 (Formal Affine Map). For every division word ω , define

$$F_{\omega} : \mathbb{Q} \rightarrow \mathbb{Q}$$

by

$$F_{\omega}(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}.$$

The rational domain is used because an arbitrary formal division word need not prescribe powers of 2 that divide the corresponding numerators.

For the empty word,

$$F_{\emptyset}(x) = x,$$

since

$$|\emptyset| = D(\emptyset) = c(\emptyset) = 0.$$

Theorem 4.1 (Affine Representation). *For every division word ω and every $x \in \mathbb{Q}$,*

$$S_{\omega}(x) = F_{\omega}(x).$$

Equivalently, symbolic composition of the prescribed steps has the closed affine form

$$S_{\omega}(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}.$$

Proof. We argue by induction on the word.

For the empty word,

$$S_{\emptyset}(x) = x = F_{\emptyset}(x).$$

Now let

$$\omega = (d_0, \eta),$$

where η is the tail of the word. By the induction hypothesis,

$$S_{\eta}(y) = \frac{3^{|\eta|}y + c(\eta)}{2^{D(\eta)}}$$

for every $y \in \mathbb{Q}$. Therefore

$$\begin{aligned} S_{\omega}(x) &= S_{\eta}\left(\frac{3x+1}{2^{d_0}}\right) \\ &= \frac{3^{|\eta|}\left(\frac{3x+1}{2^{d_0}}\right) + c(\eta)}{2^{D(\eta)}} \\ &= \frac{3^{|\eta|+1}x + 3^{|\eta|} + 2^{d_0}c(\eta)}{2^{d_0+D(\eta)}}. \end{aligned}$$

Since

$$|\omega| = |\eta| + 1, \quad D(\omega) = d_0 + D(\eta),$$

and

$$c(\omega) = 3^{|\eta|} + 2^{d_0}c(\eta),$$

we obtain

$$S_{\omega}(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}} = F_{\omega}(x).$$

□

Corollary 4.2 (Affine Representation of a Realized Word). *If $x \in \mathbb{N}$ realizes a division word ω , then*

$$F_{\omega}(x) = f^{|\omega|}(x).$$

Proof. Write

$$\omega = (d_0, \dots, d_{m-1}).$$

Since x realizes ω ,

$$d_i = v_2(3f^i(x) + 1), \quad 0 \leq i < m.$$

Consequently, at each stage the prescribed symbolic step agrees with the corresponding application of the arithmetic map f :

$$S_{d_i}(f^i(x)) = f^{i+1}(x).$$

Iterating through the word gives

$$S_\omega(x) = f^m(x).$$

The affine representation theorem then yields

$$F_\omega(x) = S_\omega(x) = f^{|\omega|}(x).$$

□

Remark (Formal Integrality and Exact Realization). The denominator in the affine representation is

$$2^{D(\omega)}.$$

Thus, for $x \in \mathbb{N}$, the formal affine value $F_\omega(x)$ is an integer precisely when

$$3^{|\omega|}x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

This condition expresses integrality of the formal affine expression only. It does not by itself assert that x realizes ω . Exact realizability requires the prescribed powers of 2 to be the exact successive 2-adic valuations, a stronger condition developed below.

4.2 Concatenation and Prefix Structure

Let $\omega\eta$ denote the concatenation obtained by appending the division word η to the division word ω . Then

$$|\omega\eta| = |\omega| + |\eta|$$

and

$$D(\omega\eta) = D(\omega) + D(\eta).$$

Proposition 4.1 (Concatenation of Symbolic and Affine Actions). For all division words ω and η ,

$$S_{\omega\eta} = S_\eta \circ S_\omega$$

and consequently

$$F_{\omega\eta} = F_\eta \circ F_\omega.$$

Proof. The first identity follows directly from the definition of symbolic action: the entries of ω are applied first, followed by the entries of η .

By the affine representation theorem,

$$S_\omega = F_\omega, \quad S_\eta = F_\eta, \quad S_{\omega\eta} = F_{\omega\eta}.$$

Therefore

$$F_{\omega\eta} = S_{\omega\eta} = S_\eta \circ S_\omega = F_\eta \circ F_\omega.$$

□

Corollary 4.3 (Affine Constant under Concatenation). *For all division words ω and η ,*

$$c(\omega\eta) = 3^{|\eta|}c(\omega) + 2^{D(\omega)}c(\eta).$$

Proof. By the preceding proposition,

$$F_{\omega\eta}(x) = F_\eta(F_\omega(x)).$$

Using the affine representations of F_ω and F_η ,

$$\begin{aligned} F_\eta(F_\omega(x)) &= \frac{3^{|\eta|} \left(\frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}} \right) + c(\eta)}{2^{D(\eta)}} \\ &= \frac{3^{|\omega|+|\eta|}x + 3^{|\eta|}c(\omega) + 2^{D(\omega)}c(\eta)}{2^{D(\omega)+D(\eta)}}. \end{aligned}$$

On the other hand,

$$F_{\omega\eta}(x) = \frac{3^{|\omega|+|\eta|}x + c(\omega\eta)}{2^{D(\omega)+D(\eta)}}.$$

Comparing the constant terms gives

$$c(\omega\eta) = 3^{|\eta|}c(\omega) + 2^{D(\omega)}c(\eta).$$

□

For a division word ω , let

$$R_\omega := \{x \in \mathbb{N} : x \text{ realizes } \omega\}$$

denote its realization set.

Proposition 4.2 (Prefix Nesting). For all division words ω and η ,

$$R_{\omega\eta} \subseteq R_\omega.$$

Equivalently, if ω is a prefix of a division word ν , then

$$R_\nu \subseteq R_\omega.$$

Proof. Suppose $x \in R_{\omega\eta}$. Then the initial $|\omega| + |\eta|$ division counts generated by x are precisely the entries of $\omega\eta$. Their first $|\omega|$ entries are therefore the entries of ω . Hence x realizes ω , and so

$$x \in R_\omega.$$

Thus

$$R_{\omega\eta} \subseteq R_\omega.$$

The equivalent prefix formulation follows because any extension of ω can be written as $\omega\eta$ for some division word η . $\square$

Remark. Concatenation therefore has parallel algebraic and dynamical meanings:

$$\omega \longmapsto \omega\eta$$

corresponds algebraically to composition of affine maps,

$$F_{\omega\eta} = F_\eta \circ F_\omega,$$

and dynamically to refinement of realization sets,

$$R_{\omega\eta} \subseteq R_\omega.$$

The latter nesting principle will become the arithmetic nesting of exact dyadic realization classes once realizability is characterized congruentially.

4.3 Explicit Form of the Affine Constant

The recursive definition of $c(\omega)$ is convenient for constructing the affine map, but it obscures the contribution of the individual additive terms introduced along the symbolic trajectory. We now make those contributions explicit.

Let

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word. For $0 \leq j \leq m$, define the *prefix division count*

$$D_j(\omega) := \sum_{i=0}^{j-1} d_i.$$

Thus

$$D_0(\omega) = 0, \quad D_m(\omega) = D(\omega).$$

Theorem 4.4 (Explicit Form of the Affine Constant). *For every division word*

$$\omega = (d_0, \dots, d_{m-1}),$$

the affine constant is

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{D_j(\omega)}.$$

Equivalently,

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{\sum_{i=0}^{j-1} d_i},$$

where the sum in the exponent is 0 when $j = 0$.

Proof. We argue by induction on the length m .

For $m = 0$, the word is empty. Both the defining value

$$c(\emptyset) = 0$$

and the displayed sum are zero.

Now suppose $m \geq 1$, and write

$$\omega = (d_0, \eta), \quad \eta = (d_1, \dots, d_{m-1}).$$

By the recursive definition of the affine constant,

$$c(\omega) = 3^{m-1} + 2^{d_0} c(\eta).$$

Applying the induction hypothesis to the tail η gives

$$c(\eta) = \sum_{k=0}^{m-2} 3^{m-2-k} 2^{\sum_{i=1}^k d_i},$$

where the exponent sum is empty when $k = 0$. Hence

$$c(\omega) = 3^{m-1} + \sum_{k=0}^{m-2} 3^{m-2-k} 2^{d_0 + \sum_{i=1}^k d_i}.$$

Set $j = k + 1$. Then

$$m - 2 - k = m - 1 - j$$

and

$$d_0 + \sum_{i=1}^k d_i = \sum_{i=0}^{j-1} d_i = D_j(\omega).$$

The initial term is

$$3^{m-1} = 3^{m-1} 2^{D_0(\omega)},$$

so adjoining it as the $j = 0$ term yields

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{D_j(\omega)}.$$

□

Corollary 4.5. *For every division word $\omega = (d_0, \dots, d_{m-1})$,*

$$F_\omega(x) = \frac{3^m x + \sum_{j=0}^{m-1} 3^{m-1-j} 2^{D_j(\omega)}}{2^{D(\omega)}}.$$

Remark. The explicit formula records the accumulated contribution of each additive +1 introduced during symbolic iteration. The contribution created at step j is multiplied by

$$3^{m-1-j}$$

during the remaining symbolic steps, while the factor

$$2^{D_j(\omega)}$$

records all prescribed divisions that occurred before that contribution was introduced.

Thus the affine constant retains the time ordering of the division word, not merely its total division count.

4.4 Dyadic-Ternary Balance

The affine representation

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}$$

contains two cumulative multiplicative factors: the ternary factor $3^{|\omega|}$, contributed by the $3x + 1$ term at each symbolic step, and the dyadic factor $2^{D(\omega)}$, arising from the prescribed divisions.

Their balance is governed by a natural integer threshold.

Definition 4.5 (Dyadic Threshold). For $m \in \mathbb{N}$, let $B(m)$ be the least natural number k such that

$$3^m < 2^k.$$

Thus

$$B(m) = \min\{k \in \mathbb{N} : 3^m < 2^k\}.$$

Since $0 \in \mathbb{N}$, the definition includes $m = 0$. In particular,

$$B(0) = 1,$$

since $3^0 = 1 < 2$, while $1 \not< 1$.

Lemma 4.6 (Threshold Characterization). For all $m, k \in \mathbb{N}$,

$$B(m) \leq k \iff 3^m < 2^k.$$

Proof. By definition, $B(m)$ is the least exponent for which

$$3^m < 2^{B(m)}.$$

If $B(m) \leq k$, monotonicity of 2^k gives

$$3^m < 2^{B(m)} \leq 2^k.$$

Conversely, if $3^m < 2^k$, then k belongs to the defining set for $B(m)$, and minimality gives

$$B(m) \leq k.$$

□

Proposition 4.3 (Logarithmic Form of the Threshold). For every $m \in \mathbb{N}$,

$$B(m) = \lfloor m \log_2 3 \rfloor + 1.$$

Proof. The inequality

$$3^m < 2^k$$

is equivalent to

$$m \log_2 3 < k.$$

Hence $B(m)$ is the least integer strictly greater than $m \log_2 3$, namely

$$\lfloor m \log_2 3 \rfloor + 1.$$

□

Remark. The sequence $B(m)$ is OEIS sequence A020914. The OEIS identification is descriptive only; the argument below uses the intrinsic threshold characterization.

Definition 4.6 (Affine Slope). For a division word ω , define its *affine slope* by

$$\lambda_\omega := \frac{3^{|\omega|}}{2^{D(\omega)}}.$$

Thus

$$F_\omega(x) = \lambda_\omega x + \frac{c(\omega)}{2^{D(\omega)}}.$$

Theorem 4.7 (Dyadic Threshold and Affine Slope). For every division word ω ,

$$\lambda_\omega < 1 \iff B(|\omega|) \leq D(\omega).$$

Equivalently,

$$\lambda_\omega < 1 \iff 3^{|\omega|} < 2^{D(\omega)}.$$

Proof. Since $D(\omega) \in \mathbb{N}$, we have

$$2^{D(\omega)} \geq 1,$$

and hence the denominator of λ_ω is positive. Therefore

$$\lambda_\omega < 1 \iff 3^{|\omega|} < 2^{D(\omega)}.$$

Applying the threshold characterization with

$$m = |\omega|, \quad k = D(\omega)$$

gives

$$3^{|\omega|} < 2^{D(\omega)} \iff B(|\omega|) \leq D(\omega).$$

□

Corollary 4.8 (Strict Slope Dichotomy). *If ω is nonempty, then*

$$\lambda_\omega < 1 \quad \text{or} \quad \lambda_\omega > 1.$$

In particular,

$$\lambda_\omega \neq 1.$$

Proof. Equality would require

$$3^{|\omega|} = 2^{D(\omega)}.$$

If $D(\omega) > 0$, the right-hand side is even while the left-hand side is odd. If $D(\omega) = 0$, the right-hand side is 1, whereas $3^{|\omega|} > 1$ because ω is nonempty. Thus equality is impossible. □

Corollary 4.9 (Positivity of the Affine Constant). *If ω is nonempty, then*

$$c(\omega) > 0.$$

Proof. Let $|\omega| = m \geq 1$. By the explicit formula,

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{D_j(\omega)}.$$

Every summand is nonnegative, and the $j = 0$ term is

$$3^{m-1} > 0.$$

Hence $c(\omega) > 0$. □

Remark (Slope versus Descent). The inequality

$$\lambda_\omega < 1$$

is a statement about the linear coefficient of F_ω ; it is not by itself a statement that the encoded trajectory descends below its starting value.

Indeed,

$$F_\omega(x) - x = \frac{(3^{|\omega|} - 2^{D(\omega)})x + c(\omega)}{2^{D(\omega)}}.$$

When

$$3^{|\omega|} < 2^{D(\omega)},$$

the linear contribution is negative, but the affine constant $c(\omega) > 0$ offsets part of that decrease. Thus the slope determines the eventual direction of the affine map, while actual descent also depends on the starting value and the affine constant.

The relation between subunit slope and genuine descent on realized trajectories will be established later.

4.5 Formal Integrality versus Exact Realization

Consider the division word

$$\omega = (1, 1, 1, 4).$$

It has length

$$|\omega| = 4$$

and total division count

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Since

$$B(4) = 7,$$

this word lies exactly at the dyadic threshold:

$$3^4 = 81 < 128 = 2^7,$$

and therefore

$$\lambda_\omega = \frac{81}{128} < 1.$$

The prefix division counts are

$$D_0(\omega) = 0, \quad D_1(\omega) = 1, \quad D_2(\omega) = 2, \quad D_3(\omega) = 3.$$

Hence the explicit formula for the affine constant gives

$$\begin{aligned} c(\omega) &= 3^3 2^0 + 3^2 2^1 + 3^1 2^2 + 3^0 2^3 \\ &= 27 + 18 + 12 + 8 \\ &= 65. \end{aligned}$$

Thus

$$F_{\omega}(x) = \frac{81x + 65}{128}.$$

For $F_{\omega}(x)$ to be integral, it is necessary and sufficient that

$$81x + 65 \equiv 0 \pmod{128}.$$

Since

$$81^{-1} \equiv 49 \pmod{128},$$

this is equivalent to

$$x \equiv 15 \pmod{128}.$$

This residue class therefore describes exactly the natural numbers for which the *formal* affine expression associated with ω is integral. It does not, however, follow that every such number realizes ω .

Indeed, the least positive representative $x = 15$ satisfies the integrality congruence, but its fully accelerated trajectory begins

$$15 \mapsto 23 \mapsto 35 \mapsto 53 \mapsto 5.$$

The first three division counts are 1, 1, 1, but at the fourth step

$$3 \cdot 53 + 1 = 160 = 2^5 \cdot 5.$$

Thus the fourth exact valuation is 5, whereas ω prescribes 4.

The formal symbolic step prescribed by ω would instead give

$$\frac{3 \cdot 53 + 1}{2^4} = 10.$$

Accordingly,

$$F_{\omega}(15) = 10,$$

while the actual fourth fully accelerated iterate is

$$f^4(15) = 5.$$

The formal affine expression is integral, but 15 does not realize ω .

If x did realize ω , then

$$F_{\omega}(x) = f^4(x)$$

would be odd. Hence exact realization necessarily imposes the additional parity condition

$$\frac{81x + 65}{128} \equiv 1 \pmod{2},$$

or equivalently

$$81x + 65 \equiv 128 \pmod{256}.$$

Since

$$81^{-1} \equiv 177 \pmod{256},$$

this strengthened congruence is equivalent to

$$x \equiv 143 \pmod{256}.$$

Taking the least positive representative $x = 143$, we find

$$143 \mapsto 215 \mapsto 323 \mapsto 485 \mapsto 91.$$

Indeed,

$$3 \cdot 143 + 1 = 2 \cdot 215,$$

$$3 \cdot 215 + 1 = 2 \cdot 323,$$

$$3 \cdot 323 + 1 = 2 \cdot 485,$$

$$3 \cdot 485 + 1 = 2^4 \cdot 91,$$

so the successive exact 2-adic valuations are

$$(1, 1, 1, 4).$$

Thus 143 genuinely realizes ω .

Remark (Interpretation). The example exhibits two distinct arithmetic conditions.

The congruence

$$3^{|\omega|}x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}$$

expresses only integrality of the formal affine map. For the present word, it gives the class

$$x \equiv 15 \pmod{128},$$

which contains values such as 15 that do not realize ω .

Realization necessarily requires the quotient obtained after removing the prescribed total power $2^{D(\omega)}$ to be odd. In this example that adds one dyadic bit and refines the integrality class to

$$x \equiv 143 \pmod{256}.$$

The direct calculation above shows that 143 realizes the word. It does not yet establish that the strengthened congruence characterizes all realizations of an arbitrary admissible division word. The following development proves precisely that stronger statement.

5 Exact Realizability and Residue Structure

The preceding section associated a formal affine map with every division word, independently of whether that word is realized under iteration of the arithmetic map f . We now determine the additional arithmetic conditions that distinguish formal symbolic integrality from exact realization.

5.1 Affine Integrality and Prefix Propagation

Theorem 5.1 (Natural Affine Identity for a Realized Word). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word, and let $x \in \mathbb{N}$ realize ω . Then

$$3^m x + c(\omega) = 2^{D(\omega)} f^m(x).$$

Proof. Since x realizes ω , the affine representation of a realized word gives

$$F_\omega(x) = f^m(x).$$

By definition,

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

Hence

$$\frac{3^m x + c(\omega)}{2^{D(\omega)}} = f^m(x).$$

Multiplying by $2^{D(\omega)}$ yields

$$3^m x + c(\omega) = 2^{D(\omega)} f^m(x).$$

□

Corollary 5.2 (Formal Integrality of a Realized Word). *If $x \in \mathbb{N}$ realizes a division word ω , then*

$$3^{|\omega|} x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Proof. By Theorem 5.1,

$$3^{|\omega|}x + c(\omega) = 2^{D(\omega)}f^{|\omega|}(x),$$

so the affine numerator is divisible by $2^{D(\omega)}$. □

Remark. The converse is not yet available. Divisibility of the affine numerator by $2^{D(\omega)}$ guarantees only that the formal affine endpoint is integral. It does not assert that the prescribed powers of 2 are the exact successive 2-adic valuations.

The example of $\omega = (1, 1, 1, 4)$ showed this distinction explicitly: $x = 15$ satisfies the formal integrality congruence but does not realize the word.

Theorem 5.3 (Prefix Congruence Propagation). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be a division word, let $x \in \mathbb{N}$, and suppose

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

For $0 \leq j \leq m$, let

$$\omega_{\leq j} = (d_0, \dots, d_{j-1})$$

denote the prefix of length j . Then

$$3^j x + c(\omega_{\leq j}) \equiv 0 \pmod{2^{D(\omega_{\leq j})}}.$$

Thus formal integrality of the full affine numerator forces the corresponding formal integrality condition at every prefix.

Proof. Fix j with $0 \leq j \leq m$, and write

$$\omega = \omega_{\leq j}\eta,$$

where η is the remaining suffix.

The concatenation formula for the affine constant gives

$$c(\omega) = 3^{|\eta|}c(\omega_{\leq j}) + 2^{D(\omega_{\leq j})}c(\eta),$$

and

$$m = j + |\eta|.$$

Therefore

$$3^m x + c(\omega) = 3^{|\eta|} (3^j x + c(\omega_{\leq j})) + 2^{D(\omega_{\leq j})} c(\eta).$$

Moreover,

$$D(\omega) = D(\omega_{\leq j}) + D(\eta),$$

so

$$2^{D(\omega_{\leq j})} \mid 2^{D(\omega)}.$$

The assumed congruence modulo $2^{D(\omega)}$ therefore implies

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega_{\leq j})}}.$$

The second term in the preceding decomposition is already divisible by $2^{D(\omega_{\leq j})}$, hence

$$3^{|\eta|} (3^j x + c(\omega_{\leq j})) \equiv 0 \pmod{2^{D(\omega_{\leq j})}}.$$

Since $3^{|\eta|}$ is odd, it is coprime to every power of 2 and is therefore invertible modulo $2^{D(\omega_{\leq j})}$. Cancelling this factor gives

$$3^j x + c(\omega_{\leq j}) \equiv 0 \pmod{2^{D(\omega_{\leq j})}}.$$

Since j was arbitrary, the result holds for every prefix. □

Remark (Endpoint Integrality versus Stepwise Integrality). The prefix theorem shows that the global affine congruence is not merely an endpoint condition. If

$$3^{|\omega|} x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}},$$

then for every prefix $\omega_{\leq j}$,

$$F_{\omega_{\leq j}}(x) = \frac{3^j x + c(\omega_{\leq j})}{2^{D(\omega_{\leq j})}}$$

is a natural number.

Thus the prescribed symbolic divisions are arithmetically compatible at every prefix stage. What remains unresolved is *exactness*: the quotient after a prescribed division may still contain an additional factor of 2.

Exact realization therefore requires one further parity bit. The next subsection identifies that local exactness condition and then lifts it to the complete division word.

5.2 Exact Valuation and the Additional Parity Bit

Formal integrality guarantees that the prescribed power of 2 divides the relevant numerator. Exact realization requires more: after that power of 2 has been removed, the remaining quotient must be odd.

The local arithmetic condition is elementary but fundamental.

Lemma 5.4 (Exact Valuation Congruence). *Let $x, d \in \mathbb{N}$. Then*

$$v_2(3x + 1) = d$$

if and only if

$$3x + 1 \equiv 2^d \pmod{2^{d+1}}.$$

Equivalently,

$$3x + 1 = 2^d q$$

for some odd $q \in \mathbb{N}$.

Proof. If

$$v_2(3x + 1) = d,$$

then by definition

$$3x + 1 = 2^d q$$

for some odd natural number q . Writing

$$q = 2k + 1$$

gives

$$3x + 1 = 2^d(2k + 1) = 2^{d+1}k + 2^d,$$

and hence

$$3x + 1 \equiv 2^d \pmod{2^{d+1}}.$$

Conversely, suppose

$$3x + 1 \equiv 2^d \pmod{2^{d+1}}.$$

Then

$$3x + 1 = 2^{d+1}k + 2^d = 2^d(2k + 1)$$

for some $k \in \mathbb{N}$. Since $2k + 1$ is odd, exactly d factors of 2 divide $3x + 1$. Therefore

$$v_2(3x + 1) = d.$$

□

Remark. The distinction between divisibility and exact valuation is therefore the distinction between

$$3x + 1 \equiv 0 \pmod{2^d}$$

and

$$3x + 1 \equiv 2^d \pmod{2^{d+1}}.$$

The first says only that the prescribed division by 2^d is possible. The second says that the quotient after that division is odd, and therefore that d is the exact 2-adic valuation.

5.3 The Exact Realization Criterion

The preceding lemma gives a local criterion for one exact symbolic step. The next theorem shows that, for an admissible division word, all of the local exact-valuation conditions are encoded simultaneously by one global affine congruence.

Theorem 5.5 (Exact Realization Criterion). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word, and let $x \in \mathbb{N}$. Then x realizes ω if and only if

$$3^m x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}.$$

Proof. Suppose first that x realizes ω . Since ω is admissible, it is nonempty and its first division count satisfies $d_0 \geq 1$. Since x realizes ω ,

$$d_0 = v_2(3x + 1).$$

Hence

$$v_2(3x + 1) \geq 1,$$

so x is odd. Since $x \in \mathbb{N}$, it follows that $x > 0$. Thus the realized trajectory is genuinely a fully accelerated Collatz trajectory.

By the natural affine identity,

$$3^m x + c(\omega) = 2^{D(\omega)} f^m(x).$$

Since $m \geq 1$, the accelerated endpoint $f^m(x)$ is odd. Hence

$$f^m(x) \equiv 1 \pmod{2}.$$

Multiplying this congruence by $2^{D(\omega)}$ gives

$$3^m x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}.$$

For the converse, we prove a slightly stronger statement by induction on the word length.

Let

$$\eta = (e_0, \dots, e_{n-1})$$

be any finite division word whose entries are positive, allowing also the empty word when $n = 0$. We claim that if

$$3^n x + c(\eta) \equiv 2^{D(\eta)} \pmod{2^{D(\eta)+1}},$$

then x is odd and realizes η .

If $n = 0$, then

$$D(\eta) = 0, \quad c(\eta) = 0,$$

and the exact congruence becomes

$$x \equiv 1 \pmod{2}.$$

Thus x is odd. The empty word is realized trivially, so the claim holds.

Now suppose $n \geq 1$, and write

$$\eta = (e_0, \eta'), \quad \eta' = (e_1, \dots, e_{n-1}).$$

The exact congruence implies, after reducing modulo $2^{D(\eta)}$,

$$3^n x + c(\eta) \equiv 0 \pmod{2^{D(\eta)}}.$$

By prefix congruence propagation, the one-term prefix (e_0) therefore satisfies

$$3x + 1 \equiv 0 \pmod{2^{e_0}}.$$

Hence there exists $y \in \mathbb{N}$ such that

$$3x + 1 = 2^{e_0} y.$$

Using

$$c(\eta) = 3^{n-1} + 2^{e_0} c(\eta')$$

and

$$D(\eta) = e_0 + D(\eta'),$$

we obtain

$$\begin{aligned} 3^n x + c(\eta) &= 3^{n-1}(3x + 1) + 2^{e_0}c(\eta') \\ &= 2^{e_0} (3^{n-1}y + c(\eta')) . \end{aligned}$$

Substituting this factorization into the exact congruence gives

$$2^{e_0} (3^{n-1}y + c(\eta')) \equiv 2^{e_0} 2^{D(\eta')} \pmod{2^{e_0} 2^{D(\eta') + 1}}.$$

Since the modulus itself contains the factor 2^{e_0} , this congruence is equivalent to divisibility by

$$2^{e_0} 2^{D(\eta') + 1}$$

of the difference of the two sides. Cancelling the common positive factor 2^{e_0} in that divisibility relation yields

$$3^{n-1}y + c(\eta') \equiv 2^{D(\eta')} \pmod{2^{D(\eta') + 1}}.$$

Every entry of η' is positive, so the induction hypothesis applies. Therefore y is odd and realizes η' .

Since

$$3x + 1 = 2^{e_0}y$$

with y odd, Lemma 5.4 gives

$$v_2(3x + 1) = e_0.$$

Consequently,

$$f(x) = y.$$

Moreover $e_0 \geq 1$, so $3x + 1$ is even, which forces x to be odd.

Thus the first exact division count of x is e_0 , and its next accelerated iterate $y = f(x)$ realizes the tail η' . Therefore x realizes the complete word η .

This proves the induction statement. Applying it to the admissible word ω proves the converse implication. $\square$

Remark (The Additional Dyadic Bit). The distinction between the two affine congruences is now exact.

The congruence

$$3^{|\omega|}x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}$$

expresses formal integrality. It guarantees that all prescribed divisions are arithmetically possible, including at every prefix, but does not require those divisions to remove the complete power of 2.

By contrast,

$$3^{|\omega|}x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}$$

is equivalent, for admissible ω , to exact realization of the entire division word.

The additional dyadic bit records that, after the prescribed factor $2^{D(\omega)}$ has been removed, the resulting endpoint is odd. Remarkably, together with prefix integrality this single endpoint parity condition forces exactness at every intermediate step.

5.4 Exact Realization Residue Classes

The exact realization criterion converts realization into a single linear congruence in the initial value. Because its coefficient is an odd power of 3, that congruence determines one and only one dyadic residue class.

Theorem 5.6 (Exact Realization Residue Class). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be an admissible division word. Then there exists a unique residue class

$$r_\omega \pmod{2^{D(\omega)+1}}$$

such that

$$x \text{ realizes } \omega \iff x \equiv r_\omega \pmod{2^{D(\omega)+1}}.$$

Equivalently, there is a unique canonical representative

$$0 \leq r_\omega < 2^{D(\omega)+1}$$

with this property.

Proof. By Theorem 5.5, x realizes ω if and only if

$$3^m x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}.$$

Since 3^m is odd,

$$\gcd(3^m, 2^{D(\omega)+1}) = 1.$$

Hence multiplication by 3^m is invertible modulo $2^{D(\omega)+1}$. Therefore the affine linear congruence

$$3^m x + c(\omega) \equiv 2^{D(\omega)} \pmod{2^{D(\omega)+1}}$$

has a unique solution class for x modulo $2^{D(\omega)+1}$.

Denote its unique canonical representative by r_ω . Then

$$x \text{ realizes } \omega \iff x \equiv r_\omega \pmod{2^{D(\omega)+1}},$$

as required. $\square$

Corollary 5.7 (Admissible Division Words Are Realizable). *Every admissible division word is realizable.*

Proof. Let ω be admissible. By Theorem 5.6, its exact realization congruence has a canonical solution

$$r_\omega \pmod{2^{D(\omega)+1}}.$$

Taking $x = r_\omega$ gives

$$x \equiv r_\omega \pmod{2^{D(\omega)+1}},$$

and hence x realizes ω . $\square$

Remark. Admissibility was defined only as an intrinsic positivity condition on a finite nonempty division word. We have now proved, rather than assumed, that this condition is sufficient for finite realizability.

Since ω is admissible, its first division count is positive. Therefore any realizer of ω , and in particular r_ω , is odd and positive.

Every finite nonempty sequence of positive division counts occurs
along some fully accelerated Collatz trajectory.

Moreover, its complete realization set is exactly one residue class modulo

$$2^{D(\omega)+1}.$$

The next subsection studies how these exact realization classes behave when division words are extended through their prefixes.

5.5 Nested Structure of Realization Classes

For a division word ω , recall its realization set

$$R_\omega = \{x \in \mathbb{N} : x \text{ realizes } \omega\}.$$

If ω is admissible, the preceding results show that R_ω is exactly one residue class modulo $2^{D(\omega)+1}$:

$$R_\omega = \{x \in \mathbb{N} : x \equiv r_\omega \pmod{2^{D(\omega)+1}}\},$$

where

$$0 \leq r_\omega < 2^{D(\omega)+1}$$

is uniquely determined.

We now show that the prefix structure of division words induces a corresponding nested-or-disjoint structure on their realization sets.

Lemma 5.8 (Common Realizations Force Prefix Comparability). *Let ω and η be division words. If some $x \in \mathbb{N}$ realizes both ω and η , then one word is a prefix of the other.*

More precisely, if

$$|\omega| \leq |\eta|,$$

then

$$\omega = \eta_{\leq |\omega|},$$

where $\eta_{\leq |\omega|}$ denotes the prefix of η of length $|\omega|$.

Proof. Suppose that x realizes both ω and η , and assume

$$|\omega| \leq |\eta|.$$

Since realization of a word implies realization of each of its prefixes, x realizes the prefix

$$\eta_{\leq |\omega|}.$$

Both ω and $\eta_{\leq |\omega|}$ therefore have length $|\omega|$ and are realized by the same starting value x . But a fixed starting value determines a unique division word of each finite length. Hence

$$\omega = \eta_{\leq |\omega|}.$$

Interchanging the roles of ω and η gives the opposite case. Thus any two division words having a common realization are prefix-comparable. $\square$

Theorem 5.9 (Nested-or-Disjoint Realization Classes). *Let ω and η be division words. Then at least one of*

$$R_\eta \subseteq R_\omega, \quad R_\omega \subseteq R_\eta, \quad R_\omega \cap R_\eta = \emptyset$$

holds.

Equivalently, two realization sets are either nested or disjoint; they cannot partially overlap.

Proof. If

$$R_\omega \cap R_\eta = \emptyset,$$

there is nothing to prove.

Otherwise choose

$$x \in R_\omega \cap R_\eta.$$

Then x realizes both ω and η , so by Lemma 5.8 the two words are prefix-comparable.

If ω is a prefix of η , then every natural number realizing η also realizes ω . Hence

$$R_\eta \subseteq R_\omega.$$

Similarly, if η is a prefix of ω , then

$$R_\omega \subseteq R_\eta.$$

Thus any two realization sets are nested or disjoint. □

Corollary 5.10 (Nested-or-Disjoint Exact Residue Classes). *Let ω and η be admissible division words, with exact realization residues*

$$r_\omega \pmod{2^{D(\omega)+1}} \quad \text{and} \quad r_\eta \pmod{2^{D(\eta)+1}}.$$

Then their corresponding exact residue classes are either disjoint or one is contained in the other.

Moreover, if η extends ω , then

$$r_\eta \equiv r_\omega \pmod{2^{D(\omega)+1}}.$$

The analogous statement holds with ω and η interchanged.

Proof. For admissible division words, the exact residue classes are precisely their realization sets:

$$x \equiv r_\omega \pmod{2^{D(\omega)+1}} \iff x \in R_\omega,$$

and similarly for η .

The nested-or-disjoint conclusion therefore follows immediately from Theorem 5.9.

Now suppose that η extends ω . The canonical representative r_η belongs to its own exact realization class, and hence realizes η . It therefore realizes the prefix ω , so the residue characterization of R_ω gives

$$r_\eta \equiv r_\omega \pmod{2^{D(\omega)+1}}.$$

□

Corollary 5.11 (Successive Prefix Refinement). *Let*

$$\omega = (d_0, \dots, d_{m-1})$$

be admissible, and let

$$\omega_j = (d_0, \dots, d_{j-1})$$

for $1 \leq j \leq m$. Then, whenever $1 \leq j < m$,

$$R_{\omega_{j+1}} \subseteq R_{\omega_j}$$

and

$$r_{\omega_{j+1}} \equiv r_{\omega_j} \pmod{2^{D(\omega_j)+1}}.$$

Thus each additional exact valuation condition refines the realization class of the preceding prefix.

Proof. Since ω is admissible and $1 \leq j < m$, both ω_j and ω_{j+1} are nonempty admissible prefixes of ω . Moreover, ω_{j+1} extends ω_j . Hence both conclusions follow from the preceding corollary. □

Remark. The realization classes therefore form a laminar family: extension of a division word produces a nested refinement of its realization class, whereas prefix-incomparable words have disjoint realization sets.

For admissible words these sets are not merely abstract dynamical classes; each is a single dyadic residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)+1}}.$$

The next section develops the geometry and density of such dyadic residue classes systematically by introducing them as cylinders.

6 Cylinder Geometry of Division Words

Finite admissible division words determine exact 2-adic realization classes. For each admissible division word ω , exact realizability determines a unique residue class modulo $2^{D(\omega)+1}$, with a unique canonical representative

$$0 \leq r_\omega < 2^{D(\omega)+1}.$$

These residue classes naturally belong to the more general arithmetic geometry of congruence classes modulo powers of 2. We first develop that dyadic structure independently of the Collatz dynamics and then specialize it to the realization classes associated with division words.

6.1 Dyadic Cylinders and Division-Word Cylinders

Definition 6.1 (Dyadic Cylinder). For $r, k \in \mathbb{N}$, define the dyadic cylinder

$$C(r, k) := \{x \in \mathbb{N} : x \equiv r \pmod{2^k}\}.$$

Remark. The residue r need not be canonical in this definition. When uniqueness of the representative is required, we shall impose

$$0 \leq r < 2^k.$$

The definition itself is purely arithmetic and is independent of the Collatz map.

Definition 6.2 (Cylinder of a Division Word). Let ω be an admissible division word, and let r_ω be its canonical exact realization residue. Define

$$C_\omega := C(r_\omega, D(\omega) + 1).$$

Equivalently,

$$C_\omega = \{x \in \mathbb{N} : x \equiv r_\omega \pmod{2^{D(\omega)+1}}\}.$$

By Theorem 5.6,

$$C_\omega = R_\omega = \{x \in \mathbb{N} : x \text{ realizes } \omega\}.$$

6.2 Geometry of Dyadic Cylinders

The basic geometry of dyadic cylinders is independent of the Collatz map. It follows solely from the divisibility relation among powers of 2.

Proposition 6.1 (Dyadic Cylinder Containment). Let $r, s, k, \ell \in \mathbb{N}$ with $k \leq \ell$. Then

$$C(s, \ell) \subseteq C(r, k)$$

if and only if

$$s \equiv r \pmod{2^k}.$$

Proof. Suppose first that

$$C(s, \ell) \subseteq C(r, k).$$

Since $s \in C(s, \ell)$, the inclusion implies

$$s \in C(r, k),$$

and therefore

$$s \equiv r \pmod{2^k}.$$

Conversely, suppose

$$s \equiv r \pmod{2^k},$$

and let $x \in C(s, \ell)$. Then

$$x \equiv s \pmod{2^\ell}.$$

Since $k \leq \ell$,

$$2^k \mid 2^\ell,$$

so reduction modulo 2^k gives

$$x \equiv s \pmod{2^k}.$$

Combining this with

$$s \equiv r \pmod{2^k}$$

yields

$$x \equiv r \pmod{2^k}.$$

Hence $x \in C(r, k)$, proving

$$C(s, \ell) \subseteq C(r, k).$$

□

Corollary 6.1 (Nested-or-Disjoint Dyadic Cylinders). *Any two dyadic cylinders are either nested or disjoint. More precisely, for arbitrary $r, s, k, \ell \in \mathbb{N}$, at least one of*

$$C(s, \ell) \subseteq C(r, k), \quad C(r, k) \subseteq C(s, \ell), \quad C(r, k) \cap C(s, \ell) = \emptyset$$

holds.

Proof. Assume without loss of generality that $k \leq \ell$. If the two cylinders are disjoint, there is nothing to prove.

Otherwise choose

$$x \in C(r, k) \cap C(s, \ell).$$

Then

$$x \equiv r \pmod{2^k}$$

and, after reducing the second congruence modulo 2^k ,

$$x \equiv s \pmod{2^k}.$$

Hence

$$s \equiv r \pmod{2^k}.$$

Proposition 6.1 therefore gives

$$C(s, \ell) \subseteq C(r, k).$$

The case $\ell \leq k$ is symmetric. □

Theorem 6.2 (Binary Refinement of a Dyadic Cylinder). *Let*

$$0 \leq r < 2^k.$$

Then

$$C(r, k) = C(r, k+1) \sqcup C(r+2^k, k+1),$$

where $\sqcup$ denotes disjoint union.

Proof. A residue modulo 2^{k+1} reducing to r modulo 2^k has exactly one of the two forms

$$r \quad \text{or} \quad r + 2^k.$$

Therefore every element of $C(r, k)$ lies in exactly one of

$$C(r, k+1) \quad \text{and} \quad C(r+2^k, k+1).$$

Conversely, both of these residues are congruent to r modulo 2^k , so both child cylinders are contained in $C(r, k)$.

The two children are disjoint because

$$r \not\equiv r + 2^k \pmod{2^{k+1}}.$$

Hence their union is disjoint and equals the parent cylinder. $\square$

Remark. Thus the complete family of dyadic cylinders forms a binary refinement geometry. Passing from depth k to depth $k+1$ resolves one additional binary residue bit, and any two cylinders are either nested or disjoint.

The realization cylinders introduced above inherit this arithmetic geometry after the specialization

$$r = r_\omega, \quad k = D(\omega) + 1.$$

6.3 Counting and Natural Density

For a canonical dyadic cylinder $C(r, k)$, define

$$A_{r,k}(N) := \#\{x \in \mathbb{N} : 0 \leq x < N, x \in C(r, k)\}.$$

With the convention $0 \in \mathbb{N}$, this definition also treats the zero residue class uniformly. In particular, the initial block

$$[0, 2^k)$$

contains exactly one representative of every residue class modulo 2^k , including 0. Cylinder membership is periodic with period 2^k . Moreover, if

$$0 \leq r < 2^k,$$

then every complete block of 2^k consecutive natural numbers contains exactly one member of $C(r, k)$.

Theorem 6.3 (Dyadic Cylinder Counting Bounds). *Let*

$$0 \leq r < 2^k.$$

Then for every $N \in \mathbb{N}$,

$$\left\lfloor \frac{N}{2^k} \right\rfloor \leq A_{r,k}(N) \leq \left\lfloor \frac{N}{2^k} \right\rfloor + 1.$$

Proof. Write

$$N = q 2^k + s, \quad 0 \leq s < 2^k.$$

The first $q 2^k$ natural numbers consist of q complete periods of length 2^k , each containing exactly one member of the cylinder. They therefore contribute exactly q points.

The remaining interval has length $s < 2^k$, so it contains at most one additional member of the same residue class. Hence

$$q \leq A_{r,k}(N) \leq q + 1.$$

Since

$$q = \left\lfloor \frac{N}{2^k} \right\rfloor,$$

the result follows. □

Theorem 6.4 (Natural Density of a Dyadic Cylinder). *Every canonical dyadic cylinder $C(r, k)$ has ordinary natural density*

$$2^{-k}.$$

More precisely, for $N > 0$,

$$\left| \frac{A_{r,k}(N)}{N} - \frac{1}{2^k} \right| \leq \frac{1}{N},$$

and therefore

$$\lim_{N \rightarrow \infty} \frac{A_{r,k}(N)}{N} = \frac{1}{2^k}.$$

Proof. Write

$$N = q 2^k + s, \quad 0 \leq s < 2^k.$$

By Theorem 6.3,

$$A_{r,k}(N) \in \{q, q + 1\}.$$

In either case,

$$\left| A_{r,k}(N) - \frac{N}{2^k} \right| \leq 1.$$

Dividing by $N > 0$ gives

$$\left| \frac{A_{r,k}(N)}{N} - \frac{1}{2^k} \right| \leq \frac{1}{N}.$$

Since $1/N \rightarrow 0$, the stated density follows. □

Corollary 6.5 (Density of a Realization Cylinder). *Let ω be an admissible division word. Then*

$$C_\omega = C(r_\omega, D(\omega) + 1)$$

has ordinary natural density

$$\boxed{\text{dens}(C_\omega) = 2^{-(D(\omega)+1)}}.$$

Proof. The canonical realization residue satisfies

$$0 \leq r_\omega < 2^{D(\omega)+1}.$$

Apply Theorem 6.4 with

$$r = r_\omega, \quad k = D(\omega) + 1.$$

□

6.4 Realization Cylinder Forest

We now specialize the generic dyadic geometry to the cylinders determined by admissible division words.

Write

$$\omega \preceq \eta$$

when ω is a prefix of η .

Lemma 6.6 (Proper Prefixes Increase Dyadic Depth). *Let ω be a proper prefix of an admissible division word η . Then*

$$D(\omega) < D(\eta).$$

Proof. Since ω is a proper prefix of η , write

$$\eta = \omega\xi$$

with ξ nonempty. Every entry of η , and hence every entry of ξ , is positive. Therefore

$$D(\xi) > 0.$$

By additivity of total division count under concatenation,

$$D(\eta) = D(\omega) + D(\xi),$$

and hence

$$D(\omega) < D(\eta).$$

□

Theorem 6.7 (Exact Cylinder Inclusion Criterion). *Let ω and η be admissible division words. Then*

$$C_\eta \subseteq C_\omega$$

if and only if

$$\omega \preceq \eta.$$

Equivalently,

$$\boxed{\omega \preceq \eta \iff C_\eta \subseteq C_\omega.}$$

Proof. Suppose first that

$$C_\eta \subseteq C_\omega.$$

Since

$$r_\eta \in C_\eta,$$

the assumed inclusion gives

$$r_\eta \in C_\omega.$$

Thus r_η realizes both η and ω . By Lemma 5.8, the two words are prefix-comparable.

If $\omega \preceq \eta$, we are done. Suppose instead that η is a proper prefix of ω . By Lemma 6.6,

$$D(\eta) < D(\omega).$$

Now

$$r_\eta + 2^{D(\eta)+1}$$

belongs to the same cylinder C_η as r_η . By the assumed inclusion, both numbers therefore belong to C_ω . Since C_ω is a single residue class modulo $2^{D(\omega)+1}$, the two numbers are congruent to one another modulo $2^{D(\omega)+1}$. Hence

$$2^{D(\omega)+1} \mid 2^{D(\eta)+1}.$$

But

$$D(\eta) < D(\omega),$$

and therefore

$$0 < 2^{D(\eta)+1} < 2^{D(\omega)+1},$$

making this divisibility impossible. Thus the reverse prefix relation cannot be proper, and consequently

$$\omega \preceq \eta.$$

Conversely, suppose

$$\omega \preceq \eta.$$

Every natural number realizing η also realizes each prefix of η , and hence realizes ω . Since the cylinders are exactly the corresponding realization sets,

$$C_\eta \subseteq C_\omega.$$

□

Corollary 6.8 (Forest Consequences). *Let ω and η be admissible division words.*

1.

$$C_\omega = C_\eta \iff \omega = \eta.$$

2.

$$C_\eta \subsetneq C_\omega \iff \omega \text{ is a proper prefix of } \eta.$$

3. *If neither word is a prefix of the other, then*

$$C_\omega \cap C_\eta = \emptyset.$$

4. *Distinct admissible words of equal length have disjoint cylinders.*

Proof. The first two statements follow immediately from Theorem 6.7.

For the third, generic dyadic cylinders are nested or disjoint. If C_ω and C_η were not disjoint, one would be contained in the other, and Theorem 6.7 would force one word to be a prefix of the other.

The fourth statement follows because two distinct words of equal length cannot be prefix-comparable. □

Remark (Forest Geometry). Theorem 6.7 identifies symbolic ancestry exactly with arithmetic containment, with the order reversed:

$$\omega \preceq \eta \iff C_\eta \subseteq C_\omega.$$

Since admissible division words are nonempty, there is no empty-word root. The length-one admissible words therefore form the roots of a family of disjoint prefix trees, giving a natural realization-cylinder forest.

An edge of the symbolic forest appends one additional division count. Because that division count may exceed 1, such an edge need not correspond to a single binary dyadic refinement step: it may descend several dyadic levels at once. Nevertheless every proper symbolic descendant determines a strict arithmetic refinement to a cylinder of strictly greater dyadic depth.

Thus the realization forest is embedded in the universal binary geometry of dyadic cylinders, with its symbolic prefix order represented exactly by cylinder containment.

7 Affine Contraction and Descent Cylinders

The preceding sections separated three distinct structures associated with a division word: its formal affine map, its exact realization class, and the dyadic cylinder geometry of that class. We now connect the affine slope to actual descent along realized trajectories.

Recall that

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}$$

has affine slope

$$\lambda_\omega = \frac{3^{|\omega|}}{2^{D(\omega)}}.$$

The dyadic–ternary threshold established earlier gives

$$\lambda_\omega < 1 \iff D(\omega) \geq B(|\omega|).$$

A subunit slope alone does not imply

$$F_\omega(x) < x,$$

because the affine constant $c(\omega)$ is positive. The key point is that this obstruction is finite: once the initial value is sufficiently large, the negative contribution from the subunit linear coefficient dominates the fixed affine term.

7.1 Affine Contraction and Pointwise Descent

Proposition 7.1 (Descent Threshold for Subunit Slope). Let ω be a nonempty division word satisfying

$$2^{D(\omega)} > 3^{|\omega|}.$$

Then for every $x \in \mathbb{Q}$,

$$F_\omega(x) < x$$

if and only if

$$(2^{D(\omega)} - 3^{|\omega|})x > c(\omega).$$

Equivalently,

$$F_\omega(x) < x \iff x > \frac{c(\omega)}{2^{D(\omega)} - 3^{|\omega|}}.$$

Proof. Starting from

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}},$$

and using $2^{D(\omega)} > 0$, we have

$$F_\omega(x) < x$$

if and only if

$$3^{|\omega|}x + c(\omega) < 2^{D(\omega)}x.$$

Rearranging gives

$$c(\omega) < (2^{D(\omega)} - 3^{|\omega|})x.$$

Since

$$2^{D(\omega)} - 3^{|\omega|} > 0,$$

division by this positive quantity yields the equivalent threshold

$$x > \frac{c(\omega)}{2^{D(\omega)} - 3^{|\omega|}}.$$

□

Corollary 7.1 (Eventual Descent on a Realization Cylinder). *Let ω be an admissible division word with*

$$\lambda_\omega < 1.$$

Then all sufficiently large $x \in C_\omega$ satisfy

$$f^{|\omega|}(x) < x.$$

Consequently, only finitely many members of C_ω can fail to lie below their starting value at the endpoint of the trajectory segment encoded by ω .

Proof. If $x \in C_\omega$, then x realizes the admissible word ω . Hence x is odd and positive, so the iterates $f^i(x)$ are genuine fully accelerated Collatz iterates. Moreover,

$$F_\omega(x) = f^{|\omega|}(x).$$

Since $\lambda_\omega < 1$,

$$2^{D(\omega)} > 3^{|\omega|}.$$

By Proposition 7.1,

$$F_\omega(x) < x$$

for every

$$x > \frac{c(\omega)}{2^{D(\omega)} - 3^{|\omega|}}.$$

Only finitely many natural numbers lie below this fixed bound, so only finitely many members of C_ω can fail the stated descent inequality. $\square$

Remark. This is the precise distinction between affine slope contraction and pointwise descent.

The condition

$$\lambda_\omega < 1$$

depends only on the multiplicative balance

$$3^{|\omega|} \quad \text{versus} \quad 2^{D(\omega)}.$$

Pointwise descent additionally depends on the affine constant $c(\omega)$. But for a fixed subunit-slope word, that constant contributes only the finite threshold

$$\frac{c(\omega)}{2^{D(\omega)} - 3^{|\omega|}}.$$

Thus the affine term can delay descent for finitely many small realizations, but it cannot alter the asymptotic behavior of the realization cylinder. The next subsection makes this density statement precise.

7.2 Asymptotic Descent Cylinders

Let ω be an admissible division word with

$$\lambda_\omega < 1.$$

Define its descending realization set by

$$C_\omega^\downarrow := \{x \in C_\omega : f^{|\omega|}(x) < x\}.$$

Since every $x \in C_\omega$ realizes ω ,

$$f^{|\omega|}(x) = F_\omega(x),$$

and hence equivalently

$$C_\omega^\downarrow = \{x \in C_\omega : F_\omega(x) < x\}.$$

Let

$$E_\omega := C_\omega \setminus C_\omega^\downarrow$$

denote the corresponding set of non-descending realizations.

Lemma 7.2 (Finite Affine Exceptional Set). *Let ω be admissible and satisfy*

$$\lambda_\omega < 1.$$

Then every $x \in E_\omega$ satisfies

$$(2^{D(\omega)} - 3^{|\omega|})x \leq c(\omega).$$

In particular,

$$x \leq c(\omega),$$

so E_ω is finite.

Proof. If $x \in E_\omega$, then

$$F_\omega(x) \geq x.$$

By Proposition 7.1, this is equivalent to

$$(2^{D(\omega)} - 3^{|\omega|})x \leq c(\omega).$$

Since $\lambda_\omega < 1$,

$$2^{D(\omega)} - 3^{|\omega|} > 0.$$

The difference is a positive integer, and therefore at least 1. Hence

$$x \leq (2^{D(\omega)} - 3^{|\omega|})x \leq c(\omega).$$

Thus

$$E_\omega \subseteq \{0, 1, \dots, c(\omega)\},$$

and E_ω is finite. □

Theorem 7.3 (Density Preservation under Descent). *Let ω be an admissible division word satisfying*

$$\lambda_\omega < 1.$$

Then

$$\text{dens}(C_\omega^\downarrow) = \text{dens}(C_\omega) = 2^{-(D(\omega)+1)}.$$

Proof. By Corollary 6.5,

$$\text{dens}(C_\omega) = 2^{-(D(\omega)+1)}.$$

By Lemma 7.2, the exceptional set E_ω is finite. Hence

$$\frac{\#(E_\omega \cap [0, N))}{N} \longrightarrow 0.$$

Since

$$C_\omega = C_\omega^\downarrow \sqcup E_\omega,$$

for every N ,

$$\#(C_\omega \cap [0, N)) = \#(C_\omega^\downarrow \cap [0, N)) + \#(E_\omega \cap [0, N)).$$

Dividing by N and passing to the limit gives

$$\text{dens}(C_\omega^\downarrow) = \text{dens}(C_\omega) = 2^{-(D(\omega)+1)}.$$

□

Remark. This obstruction is asymptotically negligible: the non-descending exceptional set E_ω is finite, so removing it does not change natural density:

$$\text{dens}(C_\omega^\downarrow) = \text{dens}(C_\omega).$$

Thus every admissible subunit-slope cylinder contributes its full dyadic density

$$2^{-(D(\omega)+1)}$$

to accelerated descent, up to a finite exceptional set.

Remark (From Local Descent to Global Residual Structure). The preceding theorem is a statement about a fixed admissible division word. Whenever

$$\lambda_\omega < 1,$$

the corresponding realization cylinder contributes its full natural density to accelerated descent, up to a finite exceptional set.

This is not yet a global density statement. There are infinitely many admissible division words at every positive symbolic length, so the argument cannot proceed by counting all admissible cylinders and comparing their number with the density of a typical cylinder.

Instead, the global argument must keep track of those trajectories whose prefixes have not yet entered the subunit-slope regime. The next section introduces the resulting first-contraction and residual cylinder structure. Its purpose is to partition the cylinder forest according to the first prefix at which affine contraction occurs and to estimate the dyadic mass that remains unresolved after successive refinements.

8 First-Contraction and the Residual Forest

The preceding section showed that an admissible division word with subunit affine slope contributes its full realization-cylinder density to accelerated descent, apart from finitely many exceptional initial values. To obtain a global statement, it remains to organize trajectories according to the first prefix at which the affine slope becomes strictly less than 1.

This requires a restricted symbolic structure. The full family of admissible division words is infinite at every positive word length and therefore does not provide a finite horizontal decomposition. Instead, we retain only those prefixes whose total division count has not yet reached the dyadic threshold $B(j)$. These prefixes form the *residual forest*.

For a division word ω and $1 \leq j \leq |\omega|$, write

$$\omega_{\leq j}$$

for its prefix of length j . Recall that

$$B(j)$$

is the least integer k satisfying

$$2^k > 3^j.$$

Thus

$$D(\omega_{\leq j}) \geq B(j)$$

is equivalent to the affine slope of the prefix $\omega_{\leq j}$ being strictly less than 1.

8.1 First-Contraction and Residual Words

Definition 8.1 (Residual Word). An admissible division word ω is called *residual* if every nonempty prefix has total division count strictly below the dyadic threshold:

$$D(\omega_{\leq j}) < B(j), \quad 1 \leq j \leq |\omega|.$$

Equivalently, every nonempty prefix has affine slope strictly greater than 1.

Definition 8.2 (First-Contraction Word). An admissible division word ω of length m is called a *first-contraction word* if

$$D(\omega) \geq B(m),$$

while every proper nonempty prefix remains residual:

$$D(\omega_{\leq j}) < B(j), \quad 1 \leq j < m.$$

Thus the total division count of ω reaches the dyadic threshold for the first time at its terminal symbolic position.

Lemma 8.1 (Residual Prefix Closure). *Every nonempty prefix of a residual word is residual.*

Proof. Let ω be residual and let

$$\eta = \omega_{\leq j}, \quad 1 \leq j \leq |\omega|.$$

Admissibility is inherited by nonempty prefixes. Moreover, for every $1 \leq k \leq j$,

$$\eta_{\leq k} = \omega_{\leq k}.$$

Since ω is residual,

$$D(\omega_{\leq k}) < B(k).$$

Hence

$$D(\eta_{\leq k}) < B(k)$$

for every nonempty prefix of η , so η is residual. □

Proposition 8.1 (First-Contraction Antichain). Distinct first-contraction words are prefix-incomparable. Consequently, their realization cylinders are pairwise disjoint, even when the words have different symbolic lengths.

Proof. Suppose that distinct first-contraction words ω and η satisfied

$$\omega \prec \eta,$$

where $\prec$ denotes the proper-prefix relation.

Since ω is a first-contraction word,

$$D(\omega) \geq B(|\omega|).$$

But ω is a proper prefix of the first-contraction word η , so the defining residual-prefix condition for η requires

$$D(\omega) < B(|\omega|),$$

a contradiction.

Thus neither word can be a proper prefix of the other. By the symbolic–arithmetic correspondence for realization cylinders, prefix-incomparable division words have disjoint cylinders. $\square$

Remark. The antichain property is important, but it does not by itself license an infinite sum of cylinder densities. The family of first-contraction words at a fixed symbolic length is generally infinite, since the total division count at the terminal depth may exceed the dyadic threshold at that length by an arbitrarily large amount.

The finite objects used below are instead the *residual levels*.

8.2 Residual Levels and Exit Layers

For $m \geq 1$, define the residual word level

$$\mathcal{W}_m^{\text{res}} := \{\omega : |\omega| = m \text{ and } \omega \text{ is residual}\}.$$

Proposition 8.2 (Finiteness of Residual Levels). For every $m \geq 1$, the set

$$\mathcal{W}_m^{\text{res}}$$

is finite.

Proof. If

$$\omega = (d_0, \dots, d_{m-1}) \in \mathcal{W}_m^{\text{res}},$$

then residuality at the terminal prefix gives

$$D(\omega) < B(m).$$

Since every d_i is positive and

$$d_i \leq D(\omega),$$

we obtain

$$1 \leq d_i < B(m)$$

for every i .

Thus every residual word of length m is a length- m word over the finite alphabet

$$\{1, 2, \dots, B(m) - 1\}.$$

Only finitely many such words exist. □

Corollary 8.2 (Root of the Residual Forest). *The first residual level consists of the single word*

$$\mathcal{W}_1^{\text{res}} = \{(1)\}.$$

Proof. A residual word of length one has the form (d) with $d \geq 1$, while residuality requires

$$d < B(1) = 2.$$

Hence $d = 1$. □

The first-contraction words of length one are therefore precisely the one-symbol words (d) with $d \geq 2$. Their realizations lie outside the initial residual set $\mathcal{R}_1$, so the residual-forest decomposition below begins with the exit layer $\mathcal{F}_2$. The complementary length-one population is handled separately in the final descent argument.

For $m \geq 1$, define the residual realization set

$$\mathcal{R}_m := \bigcup_{\omega \in \mathcal{W}_m^{\text{res}}} C_\omega.$$

Thus $\mathcal{R}_m$ consists precisely of starting values whose first m division counts remain residual.

Theorem 8.3 (Nested Residual Forest). *For every $m \geq 1$,*

$$\mathcal{R}_{m+1} \subseteq \mathcal{R}_m.$$

Proof. Let $x \in \mathcal{R}_{m+1}$. Then x realizes some residual word ω of length $m + 1$.

By Lemma 8.1, the prefix

$$\omega_{\leq m}$$

is residual. Since realization of a word implies realization of each of its prefixes, x realizes $\omega_{\leq m}$. Therefore

$$x \in \mathcal{R}_m.$$

□

The set of starting values whose realized trajectories leave the residual forest at the next refinement step is now defined by

$$\mathcal{F}_{m+1} := \mathcal{R}_m \setminus \mathcal{R}_{m+1}, \quad m \geq 1.$$

Theorem 8.4 (Exit Layer Equals First Contraction). *For every $m \geq 1$,*

$$\mathcal{F}_{m+1} = \{x \in \mathbb{N} : x \text{ realizes a first-contraction word of length } m + 1\}.$$

Proof. Suppose first that

$$x \in \mathcal{F}_{m+1}.$$

Then

$$x \in \mathcal{R}_m \quad \text{but} \quad x \notin \mathcal{R}_{m+1}.$$

Hence x realizes a residual word ω of length m . In particular, ω is admissible, so x is odd and positive.

Let η denote the actual division word of length $m + 1$ generated by the trajectory of x . By Lemma 3.1, η is admissible. Its length- m prefix must equal ω , because a fixed starting value determines a unique realized word of each length.

Therefore every proper nonempty prefix of η is residual. If the full word η also satisfied

$$D(\eta) < B(m + 1),$$

then η itself would be residual, and x would belong to $\mathcal{R}_{m+1}$, contrary to assumption. Hence

$$D(\eta) \geq B(m + 1),$$

so η is a first-contraction word.

Conversely, suppose that x realizes a first-contraction word η of length $m + 1$. Its length- m prefix is residual, so

$$x \in \mathcal{R}_m.$$

If $x \in \mathcal{R}_{m+1}$, then x would also realize some residual word ρ of length $m + 1$. But two words of the same length realized by the same starting value must coincide, so

$$\rho = \eta.$$

This is impossible, because ρ is residual at its terminal depth while η reaches the contraction threshold there. Thus

$$x \notin \mathcal{R}_{m+1},$$

and hence

$$x \in \mathcal{F}_{m+1}.$$

□

Corollary 8.5 (Finite-Stage Forest Decomposition). *For every $m \geq 1$,*

$$\boxed{\mathcal{R}_m = \mathcal{F}_{m+1} \sqcup \mathcal{R}_{m+1}.}$$

Proof. By definition,

$$\mathcal{F}_{m+1} = \mathcal{R}_m \setminus \mathcal{R}_{m+1},$$

and Theorem 8.3 gives

$$\mathcal{R}_{m+1} \subseteq \mathcal{R}_m.$$

The two sets are therefore disjoint and their union is exactly $\mathcal{R}_m$. □

Remark. This decomposition is the structural conservation law of the residual forest. Every trajectory that is residual through depth m has exactly one of two fates at the next symbolic step:

first contraction at depth $m + 1$,

or

continued residual survival through depth $m + 1$.

No trajectory is lost and none is counted twice.

Definition 8.3 (Infinite Residual Set). Define

$$\mathcal{R}_\infty := \bigcap_{m \geq 1} \mathcal{R}_m.$$

Thus $\mathcal{R}_\infty$ consists of those starting values whose realized division-word prefixes remain residual at every finite depth.

Corollary 8.6 (Global First-Contraction Decomposition). *The initial residual set decomposes disjointly as*

$$\mathcal{R}_1 = \left(\bigsqcup_{m \geq 1} \mathcal{F}_{m+1} \right) \sqcup \mathcal{R}_\infty.$$

Proof. Let $x \in \mathcal{R}_1$. Either

$$x \in \mathcal{R}_m$$

for every $m \geq 1$, in which case

$$x \in \mathcal{R}_\infty,$$

or else there is a least $m \geq 1$ such that

$$x \notin \mathcal{R}_{m+1}.$$

By minimality,

$$x \in \mathcal{R}_m,$$

and hence

$$x \in \mathcal{R}_m \setminus \mathcal{R}_{m+1} = \mathcal{F}_{m+1}.$$

Conversely, every exit layer $\mathcal{F}_{m+1}$ lies in $\mathcal{R}_1$ by nestedness, as does $\mathcal{R}_\infty$.

The exit layers are pairwise disjoint because an element has at most one first level at which it leaves the residual forest, and no exit-layer element belongs to $\mathcal{R}_\infty$. Hence the decomposition is disjoint. $\square$

Remark (Symbolic Conservation). Every trajectory in the initial residual set has exactly one of two possibilities: it reaches a first-contraction layer at some finite symbolic depth, or it remains residual at every finite depth.

Thus symbolic refinement neither creates nor duplicates trajectories. It partitions the initial residual set into first-contraction exits at finite symbolic depths and the infinite-depth residual set.

8.3 Residual Mass and Finite-Stage Conservation

Because $\mathcal{W}_m^{\text{res}}$ is finite and distinct words at the same symbolic length have disjoint realization cylinders, define the residual mass at depth m by

$$M_m := \sum_{\omega \in \mathcal{W}_m^{\text{res}}} 2^{-(D(\omega)+1)}.$$

Each term is exactly the natural density of the corresponding realization cylinder C_ω .

Theorem 8.7 (Residual Mass Equals Residual Density). *For every $m \geq 1$,*

$$\boxed{\text{dens}(\mathcal{R}_m) = M_m.}$$

Proof. For $N \geq 1$, let

$$A_m(N) := \#(\mathcal{R}_m \cap [0, N)).$$

Since the cylinders

$$\{C_\omega : \omega \in \mathcal{W}_m^{\text{res}}\}$$

form a finite pairwise-disjoint family,

$$A_m(N) = \sum_{\omega \in \mathcal{W}_m^{\text{res}}} A_{r_\omega, D(\omega)+1}(N),$$

where $A_{r,k}(N)$ is the dyadic-cylinder counting function introduced earlier.

For each residual word ω ,

$$\frac{A_{r_\omega, D(\omega)+1}(N)}{N} \longrightarrow 2^{-(D(\omega)+1)}.$$

Because the residual level is finite, the limits may be summed:

$$\frac{A_m(N)}{N} \longrightarrow \sum_{\omega \in \mathcal{W}_m^{\text{res}}} 2^{-(D(\omega)+1)} = M_m.$$

Hence

$$\text{dens}(\mathcal{R}_m) = M_m.$$

□

Theorem 8.8 (Exit-Layer Density). *For every $m \geq 1$, the first-contraction layer $\mathcal{F}_{m+1}$ has natural density*

$$\boxed{\text{dens}(\mathcal{F}_{m+1}) = M_m - M_{m+1}.$$

Consequently,

$$M_{m+1} \leq M_m.$$

Proof. From Corollary 8.5,

$$\mathcal{R}_m = \mathcal{F}_{m+1} \sqcup \mathcal{R}_{m+1}.$$

Therefore, for every N ,

$$\#(\mathcal{R}_m \cap [0, N)) = \#(\mathcal{F}_{m+1} \cap [0, N)) + \#(\mathcal{R}_{m+1} \cap [0, N)).$$

Rearranging and dividing by N gives

$$\frac{\#(\mathcal{F}_{m+1} \cap [0, N))}{N} = \frac{\#(\mathcal{R}_m \cap [0, N))}{N} - \frac{\#(\mathcal{R}_{m+1} \cap [0, N))}{N}.$$

Passing to the limit and applying Theorem 8.7 yields

$$\text{dens}(\mathcal{F}_{m+1}) = M_m - M_{m+1}.$$

A natural density is nonnegative, so

$$M_{m+1} \leq M_m.$$

□

Corollary 8.9 (Finite-Stage Mass Conservation). *For $m \geq 1$ and $k \geq 1$,*

$$\boxed{\sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{m+j+1}) = M_m - M_{m+k}.$$

Equivalently,

$$M_m = \sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{m+j+1}) + M_{m+k}.$$

Proof. By Theorem 8.8,

$$\text{dens}(\mathcal{F}_{m+j+1}) = M_{m+j} - M_{m+j+1}.$$

Summing over $0 \leq j < k$ gives the telescoping identity

$$\sum_{j=0}^{k-1} (M_{m+j} - M_{m+j+1}) = M_m - M_{m+k}.$$

□

Remark (Finite-Stage Versus Asymptotic Collapse). The preceding identities are entirely finite-stage statements. They show that residual mass is redistributed exactly between successive first-contraction exit layers and the portion that remains unresolved.

No assertion has yet been made that

$$M_m \longrightarrow 0.$$

In particular, no continuity-from-above property of natural density and no countable additivity over infinitely many first-contraction cylinders is being invoked.

The remaining task is quantitative: prove directly that the residual mass M_m tends to zero. That decay estimate is developed in the next section.

9 Residual Mass Collapse

The preceding section reduced the unresolved part of the symbolic dynamics to the residual forest. At each depth m , its natural density is the finite residual mass

$$M_m = \sum_{\omega \in \mathcal{W}_m^{\text{res}}} 2^{-(D(\omega)+1)},$$

and the finite-stage conservation law gives

$$\text{dens}(\mathcal{F}_{m+1}) = M_m - M_{m+1}.$$

The remaining question is therefore quantitative:

$$\boxed{M_m \longrightarrow 0 ?}$$

A direct estimate of M_m is inconvenient because extension of a word changes its dyadic weight by a factor depending on the appended division count. We therefore introduce an auxiliary weight for which all positive extensions can be summed geometrically. The resulting one-step contraction is then transferred back to the true dyadic residual mass.

9.1 Auxiliary Weighted Residual Mass

For a residual word ω , define its auxiliary weight by

$$w(\omega) := \left(\frac{3}{8}\right)^{D(\omega)}.$$

The corresponding weighted residual mass at depth m is

$$W_m := \sum_{\omega \in \mathcal{W}_m^{\text{res}}} \left(\frac{3}{8}\right)^{D(\omega)}.$$

The choice $3/8$ is adapted to the extension structure of division words. If a positive division count d is appended to ω , then

$$D(\omega d) = D(\omega) + d,$$

and hence

$$w(\omega d) = w(\omega) \left(\frac{3}{8}\right)^d.$$

For every $m \geq 1$, all possible residual extensions lie within the finite range $1 \leq d < B(m+1)$, whose geometric weight satisfies

$$\sum_{d=1}^{B(m+1)-1} \left(\frac{3}{8}\right)^d < \sum_{d=1}^{\infty} \left(\frac{3}{8}\right)^d = \frac{3/8}{1 - 3/8} = \frac{3}{5}.$$

Thus $3/5$ is a uniform upper bound for the finite extension sum at every residual depth.

Theorem 9.1 (One-Step Weighted Contraction). *For every $m \geq 1$,*

$$W_{m+1} \leq \frac{3}{5} W_m.$$

Proof. Every residual word η of length $m + 1$ has a unique decomposition

$$\eta = \omega d,$$

where ω is its length- m prefix and $d \geq 1$ is its final division count. By residual prefix closure, ω is residual.

Because η is residual at depth $m + 1$,

$$D(\eta) < B(m + 1).$$

Since

$$D(\eta) = D(\omega) + d$$

and $D(\omega) \geq 0$, it follows that

$$1 \leq d < B(m + 1).$$

Thus the actual residual children of the words in $\mathcal{W}_m^{\text{res}}$ form a subfamily of the finite collection of extensions with

$$1 \leq d < B(m + 1).$$

Since all weights are nonnegative,

$$\begin{aligned} W_{m+1} &\leq \sum_{\omega \in \mathcal{W}_m^{\text{res}}} \sum_{d=1}^{B(m+1)-1} \left(\frac{3}{8}\right)^{D(\omega)+d} \\ &= \sum_{\omega \in \mathcal{W}_m^{\text{res}}} \left(\frac{3}{8}\right)^{D(\omega)} \sum_{d=1}^{B(m+1)-1} \left(\frac{3}{8}\right)^d \\ &\leq \frac{3}{5} \sum_{\omega \in \mathcal{W}_m^{\text{res}}} \left(\frac{3}{8}\right)^{D(\omega)} \\ &= \frac{3}{5} W_m. \end{aligned}$$

□

At depth one, the only residual word is (1). Therefore

$$W_1 = \frac{3}{8} < \frac{3}{5}.$$

Corollary 9.2 (Geometric Bound for Weighted Mass). *For every $m \geq 1$,*

$$W_m \leq \left(\frac{3}{5}\right)^m.$$

Proof. The result follows by induction from Theorem 9.1 and the bound

$$W_1 \leq \frac{3}{5}.$$

□

Remark. The weighted mass W_m is auxiliary; it is not itself the natural density of the residual level. Its purpose is to expose a uniformly subcritical extension rule. The next step transfers this geometric decay to the genuine dyadic mass M_m .

9.2 Transfer to the Dyadic Residual Mass

The exact density contribution of a realization cylinder with total division count D is

$$2^{-(D+1)}.$$

This weight admits the factorization

$$\frac{1}{2^{D+1}} = \frac{1}{2} \left(\frac{4}{3}\right)^D \left(\frac{3}{8}\right)^D.$$

Indeed,

$$\frac{4}{3} \cdot \frac{3}{8} = \frac{1}{2}.$$

The final factor is precisely the auxiliary weight used in W_m . The remaining factor can be controlled uniformly across a residual level.

If

$$\omega \in \mathcal{W}_m^{\text{res}},$$

then residuality at its terminal prefix gives

$$D(\omega) < B(m),$$

and therefore

$$D(\omega) \leq B(m) - 1.$$

Since $4/3 > 1$,

$$\left(\frac{4}{3}\right)^{D(\omega)} \leq \left(\frac{4}{3}\right)^{B(m)-1}.$$

Proposition 9.1 (Weighted Envelope for Residual Mass). For every $m \geq 1$,

$$M_m \leq \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1} W_m.$$

Consequently,

$$\boxed{0 \leq M_m \leq E_m},$$

where

$$E_m := \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1} \left(\frac{3}{5}\right)^m.$$

Proof. For each $\omega \in \mathcal{W}_m^{\text{res}}$,

$$\begin{aligned} 2^{-(D(\omega)+1)} &= \frac{1}{2} \left(\frac{4}{3}\right)^{D(\omega)} \left(\frac{3}{8}\right)^{D(\omega)} \\ &\leq \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1} \left(\frac{3}{8}\right)^{D(\omega)}. \end{aligned}$$

Summing over the finite residual level gives

$$M_m \leq \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1} W_m.$$

Applying Corollary 9.2 yields

$$M_m \leq \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1} \left(\frac{3}{5}\right)^m = E_m.$$

Nonnegativity of M_m is immediate from its definition. □

The envelope E_m contains two competing effects. The factor

$$\left(\frac{4}{3}\right)^{B(m)-1}$$

grows with symbolic depth, while

$$\left(\frac{3}{5}\right)^m$$

decays geometrically. It remains to show that the latter dominates.

9.3 Five-Step Envelope Contraction

The necessary control of the threshold $B(m)$ follows from the elementary integer inequality

$$3^5 = 243 < 256 = 2^8.$$

Lemma 9.3 (Five-Step Threshold Growth). *For every $m \geq 0$,*

$$\boxed{B(m+5) \leq B(m) + 8.}$$

Proof. By definition of $B(m)$,

$$3^m < 2^{B(m)}.$$

Multiplying this inequality by

$$3^5 < 2^8$$

gives

$$3^{m+5} < 2^{B(m)+8}.$$

Since $B(m+5)$ is the least exponent k for which

$$3^{m+5} < 2^k,$$

we conclude that

$$B(m+5) \leq B(m) + 8.$$

□

Define

$$q := \left(\frac{4}{3}\right)^8 \left(\frac{3}{5}\right)^5.$$

A direct calculation gives

$$q = \frac{65536}{84375} < 1.$$

Theorem 9.4 (Five-Step Envelope Contraction). *For every $m \geq 1$,*

$$\boxed{E_{m+5} \leq qE_m.}$$

Proof. By Lemma 9.3,

$$B(m+5) - 1 \leq B(m) - 1 + 8.$$

Therefore

$$\begin{aligned} E_{m+5} &= \frac{1}{2} \left(\frac{4}{3}\right)^{B(m+5)-1} \left(\frac{3}{5}\right)^{m+5} \\ &\leq \frac{1}{2} \left(\frac{4}{3}\right)^{B(m)-1+8} \left(\frac{3}{5}\right)^{m+5} \\ &= \left(\frac{4}{3}\right)^8 \left(\frac{3}{5}\right)^5 E_m \\ &= qE_m. \end{aligned}$$

□

Corollary 9.5 (Decay of the Residual-Mass Envelope). *The envelope satisfies*

$$\boxed{E_m \longrightarrow 0 \quad (m \rightarrow \infty).}$$

Proof. Every positive integer can be written uniquely as

$$m = r + 5k, \quad r \in \{1, 2, 3, 4, 5\}, \quad k \geq 0.$$

Fix $r \in \{1, 2, 3, 4, 5\}$. Iterating Theorem 9.4 gives

$$E_{r+5k} \leq q^k E_r.$$

Since

$$0 < q < 1,$$

we have

$$q^k E_r \longrightarrow 0,$$

and therefore

$$E_{r+5k} \longrightarrow 0$$

for each $r \in \{1, 2, 3, 4, 5\}$.

These five subsequences exhaust all positive indices. Hence

$$E_m \longrightarrow 0.$$

□

Remark. The five-step grouping is not an asymptotic approximation. It is an exact integer comparison chosen so that the possible threshold growth and the weighted-mass decay can be compared uniformly:

$$3^5 < 2^8$$

controls the former, while

$$\left(\frac{3}{5}\right)^5$$

controls the latter. Their combined multiplier is strictly below 1.

9.4 Residual Mass Collapse and First-Contraction Exhaustion

Theorem 9.6 (Residual Mass Collapse). *The residual mass tends to zero:*

$$\boxed{M_m \longrightarrow 0 \quad (m \rightarrow \infty).}$$

Equivalently,

$$\text{dens}(\mathcal{R}_m) \longrightarrow 0.$$

Proof. For every $m \geq 1$, Proposition 9.1 gives

$$0 \leq M_m \leq E_m.$$

By Corollary 9.5,

$$E_m \longrightarrow 0.$$

The squeeze theorem therefore yields

$$M_m \longrightarrow 0.$$

Since

$$M_m = \text{dens}(\mathcal{R}_m),$$

the equivalent density statement follows. □

The finite-stage conservation law from the preceding section now has an asymptotic consequence.

Corollary 9.7 (Exhaustion by First-Contraction Mass). *For every $k \geq 1$,*

$$\sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{j+2}) = M_1 - M_{k+1},$$

and therefore

$$\lim_{k \rightarrow \infty} \sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{j+2}) = M_1.$$

Proof. Finite-stage mass conservation gives

$$\sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{j+2}) = M_1 - M_{k+1}.$$

By Theorem 9.6,

$$M_{k+1} \longrightarrow 0.$$

Hence the partial sums converge to M_1 . □

Remark (Asymptotic Conservation). The residual forest therefore loses no asymptotic mass except through its first-contraction exit layers. At every finite stage,

$$M_1 = \sum_{j=0}^{k-1} \text{dens}(\mathcal{F}_{j+2}) + M_{k+1},$$

and the terminal residual term tends to zero.

Thus the finite partial sums of first-contraction layer densities asymptotically exhaust the initial residual mass M_1 . This is a statement about the limit of finite-stage density identities; it does not appeal to countable additivity of natural density.

Remark. The conclusion established here concerns affine first contraction. On each fixed subunit-slope realization cylinder, the preceding section showed that the exceptional set of non-descending realizations is finite.

This fixed-cylinder statement does not yet control an entire first-contraction layer, which may contain infinitely many terminal words. The remaining task is therefore

to obtain a uniform descent cutoff at each exit depth, combine that control with residual mass collapse, and finally transfer accelerated descent to eventual descent under the ordinary Collatz map.

10 Density-One Eventual Descent

The preceding section proved that the residual mass tends to zero:

$$M_m = \text{dens}(\mathcal{R}_m) \longrightarrow 0.$$

It remains to convert this symbolic mass collapse into genuine dynamical descent and then transfer the result from the fully accelerated map to the ordinary Collatz map.

Two issues must be addressed. First, a first-contraction cylinder may contain finitely many small realizations that do not yet descend below their starting values. Since a fixed first-contraction layer may contain infinitely many child words, these exceptional sets cannot simply be united word by word. Instead, a uniform cutoff for each entire layer is required.

Second, the residual forest describes the fully accelerated dynamics of odd positive integers. The final result concerns the ordinary Collatz map on $\mathbb{N}_{>0}$. The last part of the argument identifies the initial residual class, handles the complementary congruence classes directly, and transfers accelerated descent to ordinary Collatz descent.

10.1 Uniform Descent on First-Contraction Layers

Recall that

$$\mathcal{F}_{m+1} = \mathcal{R}_m \setminus \mathcal{R}_{m+1}$$

is precisely the set of natural numbers realizing a first-contraction word of length $m + 1$.

Let

$$\eta = \omega d$$

be such a word, where ω is its residual prefix of length m and $d \geq 1$ is its final division count. The affine concatenation formula gives

$$c(\eta) = 3c(\omega) + 2^{D(\omega)}c((d)).$$

Since

$$c((d)) = 1,$$

we obtain

$$\boxed{c(\omega d) = 3c(\omega) + 2^{D(\omega)}.$$

In particular, the affine constant of a one-symbol extension depends on its residual parent but not on the value of the appended division count. This observation supplies a uniform bound across an entire first-contraction layer.

Theorem 10.1 (Uniform Descent on a First-Contraction Layer). *For every $m \geq 1$, there exists $K_m \in \mathbb{N}$ such that*

$$x \in \mathcal{F}_{m+1}, \quad x \geq K_m$$

implies

$$f^{m+1}(x) < x.$$

Proof. The residual parent level

$$\mathcal{W}_m^{\text{res}}$$

is finite. If it is empty, then

$$\mathcal{F}_{m+1} = \emptyset,$$

and the result is immediate.

Otherwise define

$$K_m := 1 + \max_{\omega \in \mathcal{W}_m^{\text{res}}} (3c(\omega) + 2^{D(\omega)}).$$

This maximum exists because $\mathcal{W}_m^{\text{res}}$ is finite and nonempty.

Now let

$$x \in \mathcal{F}_{m+1}.$$

By Theorem 8.4, x realizes some first-contraction word

$$\eta = \omega d$$

of length $m + 1$, with residual parent

$$\omega \in \mathcal{W}_m^{\text{res}}.$$

Therefore

$$c(\eta) = 3c(\omega) + 2^{D(\omega)} < K_m.$$

Since η is a first-contraction word,

$$D(\eta) \geq B(m+1),$$

and hence

$$3^{m+1} < 2^{D(\eta)}.$$

Thus

$$2^{D(\eta)} - 3^{m+1}$$

is a positive integer and therefore is at least 1.

If $x \geq K_m$, then

$$c(\eta) < x,$$

and consequently

$$\frac{c(\eta)}{2^{D(\eta)} - 3^{m+1}} \leq c(\eta) < x.$$

By Proposition 7.1,

$$F_\eta(x) < x.$$

Because x realizes η ,

$$F_\eta(x) = f^{m+1}(x),$$

so

$$f^{m+1}(x) < x.$$

□

Corollary 10.2 (Finite Non-Descent on Each Exit Layer). *For every $m \geq 1$, the set*

$$\mathcal{E}_{m+1} := \{x \in \mathcal{F}_{m+1} : f^n(x) \geq x \text{ for every } n \geq 0\}$$

is finite.

Proof. By Theorem 10.1, every

$$x \in \mathcal{F}_{m+1}$$

with $x \geq K_m$ satisfies

$$f^{m+1}(x) < x.$$

Hence every member of $\mathcal{E}_{m+1}$ lies below K_m .

□

Remark. This uniformity is essential. A fixed first-contraction layer may contain infinitely many first-contraction words, because the total division count at the terminal depth may exceed the corresponding dyadic threshold by an arbitrarily large amount. The finiteness of the exceptional set therefore does not follow by taking an infinite union of the finite exceptional sets associated with individual cylinders.

Instead, all children at a fixed layer inherit their affine constants from the finitely many residual parents. This is what produces a single cutoff K_m for the entire layer.

10.2 Density Zero of Residual Non-Descent

Define the initially residual non-descent set

$$\mathcal{N}_{\text{res}} := \{x \in \mathcal{R}_1 : f^n(x) \geq x \text{ for every } n \geq 0\}.$$

The finite-stage forest decomposition gives, for every $k \geq 1$,

$$\mathcal{R}_1 = \left(\bigsqcup_{j=2}^{k+1} \mathcal{F}_j \right) \sqcup \mathcal{R}_{k+1}.$$

Consequently, an initially residual integer that never descends either survives to residual depth $k+1$, or belongs to one of the finitely many non-descending exceptional sets from the preceding exit layers.

Proposition 10.1 (Finite-Depth Non-Descent Cover). For every $k \geq 1$, there exists a finite set

$$E_k \subseteq \mathbb{N}$$

such that

$$\boxed{\mathcal{N}_{\text{res}} \subseteq \mathcal{R}_{k+1} \cup E_k.}$$

Proof. Set

$$E_k := \bigcup_{j=2}^{k+1} \mathcal{E}_j.$$

Each $\mathcal{E}_j$ is finite by Corollary 10.2, and the union contains only finitely many layers, so E_k is finite.

Now let

$$x \in \mathcal{N}_{\text{res}}.$$

If

$$x \in \mathcal{R}_{k+1},$$

there is nothing to prove. Otherwise, the finite-stage forest decomposition places x in some

$$\mathcal{F}_j, \quad 2 \leq j \leq k+1.$$

Since x never descends, it belongs to

$$\mathcal{E}_j \subseteq E_k.$$

□

Theorem 10.3 (Initially Residual Non-Descent Has Density Zero). *The set $\mathcal{N}_{\text{res}}$ has natural density zero:*

$$\boxed{\text{dens}(\mathcal{N}_{\text{res}}) = 0.}$$

Proof. Let

$$A(N) := \#(\mathcal{N}_{\text{res}} \cap [0, N)).$$

Fix $\varepsilon > 0$.

By residual mass collapse, choose k such that

$$M_{k+1} < \frac{\varepsilon}{2}.$$

By Proposition 10.1,

$$\mathcal{N}_{\text{res}} \subseteq \mathcal{R}_{k+1} \cup E_k,$$

where E_k is finite. Writing

$$C_k := \#E_k,$$

we obtain for every N

$$A(N) \leq \#(\mathcal{R}_{k+1} \cap [0, N)) + C_k.$$

Hence

$$\frac{A(N)}{N} \leq \frac{\#(\mathcal{R}_{k+1} \cap [0, N))}{N} + \frac{C_k}{N}.$$

The depth $k+1$ is now fixed. Since

$$\text{dens}(\mathcal{R}_{k+1}) = M_{k+1},$$

the first term tends to $M_{k+1} < \varepsilon/2$, while

$$\frac{C_k}{N} \longrightarrow 0.$$

Therefore, for all sufficiently large N ,

$$\frac{A(N)}{N} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary,

$$\frac{A(N)}{N} \longrightarrow 0.$$

□

Remark. The order of the limits is important. A finite residual depth is chosen first, using

$$M_m \longrightarrow 0.$$

Only after that depth has been fixed is the interval cutoff

$$N \longrightarrow \infty$$

taken.

Thus the argument uses neither countable additivity of natural density nor an interchange of symbolic-depth and interval-size limits.

10.3 The Initial Residual Congruence Class

The abstract initial residual population has a simple arithmetic description.

Theorem 10.4 (Initial Residual Class). *For every $x \in \mathbb{N}$,*

$$\boxed{x \in \mathcal{R}_1 \iff x \equiv 3 \pmod{4}.$$

Proof. The unique residual word of length one is

$$(1).$$

Hence

$$x \in \mathcal{R}_1$$

if and only if x realizes the one-symbol word (1), equivalently,

$$v_2(3x + 1) = 1.$$

By the exact valuation criterion,

$$3x + 1 \equiv 2 \pmod{4}.$$

This is equivalent to

$$x \equiv 3 \pmod{4}.$$

□

Thus the residual forest begins with exactly one quarter of the natural numbers. The remaining congruence classes are elementary.

Proposition 10.2 (Immediate Descent Outside the Initial Residual Class). The following hold.

1. Every positive even integer x satisfies

$$T(x) < x,$$

where T denotes the ordinary Collatz map.

2. Every $x > 1$ satisfying

$$x \equiv 1 \pmod{4}$$

satisfies

$$f(x) < x.$$

Proof. If x is positive and even, then

$$T(x) = \frac{x}{2} < x.$$

Now suppose

$$x > 1, \quad x \equiv 1 \pmod{4}.$$

Then

$$4 \mid 3x + 1,$$

so

$$v_2(3x + 1) \geq 2.$$

Consequently,

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}} \leq \frac{3x+1}{4}.$$

Since $x > 1$,

$$3x+1 < 4x,$$

and therefore

$$f(x) < x.$$

□

Remark. Among positive integers, the only value not covered by the preceding immediate-descent alternatives is 1, together with the congruence class

$$x \equiv 3 \pmod{4},$$

which is precisely the initial residual class $\mathcal{R}_1$.

In this sense, the residual-forest argument resolves the single nontrivial quarter of the positive integers left after the elementary congruence reduction.

10.4 Transfer to the Ordinary Collatz Map

Define the ordinary Collatz map

$$T : \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$$

by

$$T(x) = \begin{cases} x/2, & x \equiv 0 \pmod{2}, \\ 3x+1, & x \equiv 1 \pmod{2}. \end{cases}$$

For an odd starting value x , one fully accelerated step consists of one ordinary odd Collatz step followed by exactly

$$d(x) = v_2(3x+1)$$

successive divisions by 2. Hence

$$f(x) = T^{d(x)+1}(x).$$

Iterating this observation gives the following bridge.

Theorem 10.5 (Accelerated Iterates Occur Along the Ordinary Orbit). *Let $x \in \mathbb{N}_{>0}$ be odd. For every $n \geq 0$, there exists $m \geq 0$ such that*

$$\boxed{T^m(x) = f^n(x)}.$$

Proof. Proceed by induction on n .

For $n = 0$, take $m = 0$.

Suppose the result holds after n accelerated steps. One additional fully accelerated step from the odd value $f^n(x)$ is obtained by one ordinary odd step followed by exactly

$$d(f^n(x))$$

ordinary halving steps. Hence the endpoint $f^{n+1}(x)$ is reached after finitely many additional ordinary Collatz steps. Concatenating the two finite ordinary segments proves the result. $\square$

Corollary 10.6 (Accelerated Descent Implies Ordinary Descent). *Let $x \in \mathbb{N}_{>0}$ be odd. If*

$$f^n(x) < x$$

for some n , then

$$T^m(x) < x$$

for some m .

Proof. By Theorem 10.5, choose m such that

$$T^m(x) = f^n(x).$$

Then

$$T^m(x) < x.$$

$\square$

10.5 Density-One Descent

We can now assemble the argument.

Define the ordinary eventual-descent set

$$\mathcal{D} := \{x \in \mathbb{N}_{>0} : \exists m \geq 0, T^m(x) < x\},$$

and let

$$\mathcal{N} := \mathbb{N}_{>0} \setminus \mathcal{D}$$

denote the ordinary non-descent set.

Theorem 10.7 (Density-One Eventual Descent). *The set of positive integers admitting eventual descent under the ordinary Collatz map has natural density one:*

$$\boxed{\text{dens}(\mathcal{D}) = 1.}$$

Equivalently,

$$\boxed{\lim_{N \rightarrow \infty} \frac{\#\{x \in \mathbb{N}_{>0} : x < N, \exists m \geq 0, T^m(x) < x\}}{N} = 1.}$$

Proof. We first show that the ordinary non-descent set has density zero.

Let

$$x \in \mathcal{N}$$

with

$$x \neq 1.$$

Then $x > 1$.

The integer x cannot be even, since every positive even integer descends after one ordinary Collatz step. Hence x is odd.

It also cannot satisfy

$$x \equiv 1 \pmod{4},$$

because Proposition 10.2 would give

$$f(x) < x,$$

and Corollary 10.6 would then give ordinary Collatz descent.

Therefore

$$x \equiv 3 \pmod{4}.$$

By Theorem 10.4,

$$x \in \mathcal{R}_1.$$

Moreover, x cannot descend under any accelerated iterate. If

$$f^n(x) < x$$

for some n , then accelerated descent would again imply ordinary descent, contrary to $x \in \mathcal{N}$.

Hence

$$x \in \mathcal{N}_{\text{res}}.$$

We have therefore proved

$$\boxed{\mathcal{N} \subseteq \mathcal{N}_{\text{res}} \cup \{1\}.}$$

By Theorem 10.3,

$$\text{dens}(\mathcal{N}_{\text{res}}) = 0.$$

Moreover, for every $N \geq 1$, the inclusion above gives

$$\#(\mathcal{N} \cap [0, N)) \leq \#(\mathcal{N}_{\text{res}} \cap [0, N)) + 1.$$

Therefore

$$0 \leq \frac{\#(\mathcal{N} \cap [0, N))}{N} \leq \frac{\#(\mathcal{N}_{\text{res}} \cap [0, N))}{N} + \frac{1}{N}.$$

Both terms on the right tend to 0, so by the squeeze theorem,

$$\text{dens}(\mathcal{N}) = 0.$$

For every $N \geq 1$, descent and non-descent partition the positive integers below N :

$$\{1, 2, \dots, N-1\}.$$

Consequently,

$$\#(\mathcal{D} \cap [0, N)) + \#(\mathcal{N} \cap [0, N)) = N - 1.$$

Dividing by N gives

$$\frac{\#(\mathcal{D} \cap [0, N))}{N} = 1 - \frac{1}{N} - \frac{\#(\mathcal{N} \cap [0, N))}{N}.$$

The last two terms tend to 0. Hence

$$\text{dens}(\mathcal{D}) = 1.$$

□

Remark (Scope of the Theorem). The theorem asserts that for a set of positive integers of natural density one, some ordinary Collatz iterate is strictly smaller than the starting value:

$$\exists m, \quad T^m(x) < x.$$

It does not assert that every positive starting value has this property, nor does it assert that every Collatz trajectory reaches 1. A density-zero exceptional set may still be nonempty or infinite.

11 Conclusion

This paper develops a symbolic–arithmetic framework for finite trajectories of the fully accelerated Collatz map and uses it to prove density-one eventual descent for the ordinary Collatz map. Division words encode exact sequences of 2-adic valuations, while the affine representation

$$F_\omega(x) = \frac{3^{|\omega|}x + c(\omega)}{2^{D(\omega)}}$$

translates that symbolic data into explicit arithmetic constraints.

Exact realizability requires one additional dyadic parity condition beyond formal integrality. For every admissible division word ω , its realizations form a unique residue class modulo

$$2^{D(\omega)+1}.$$

These realization classes therefore form dyadic cylinders of natural density

$$2^{-(D(\omega)+1)},$$

with symbolic prefix extension corresponding exactly to cylinder containment.

The affine slope separates symbolic prefixes according to whether it remains above 1 or has become subunit. This leads to the residual forest: residual words are those whose nonempty prefixes remain above unit affine slope, while first-contraction words cross below unit slope for the first time at their terminal position. At each finite depth the residual family is finite, and the corresponding realization sets satisfy the exact decomposition

$$\mathcal{R}_m = \mathcal{F}_{m+1} \sqcup \mathcal{R}_{m+1}.$$

Thus the unresolved population is partitioned without loss or duplication between first-contraction exits and continued residual survival.

The quantitative core of the argument is the decay of the residual mass. An auxiliary weight

$$\left(\frac{3}{8}\right)^{D(\omega)}$$

gives a uniform one-step contraction bound $3/5$. Comparison with the true dyadic cylinder mass, together with the integer threshold estimate

$$B(m+5) \leq B(m) + 8,$$

gives a strict contraction of the residual-mass envelope over five steps. Consequently,

$$M_m = \text{dens}(\mathcal{R}_m) \longrightarrow 0.$$

Affine contraction alone does not force every realization of a fixed word to descend, because the affine constant may obstruct descent for finitely many small starting values. The residual forest resolves this uniformly: at each first-contraction depth, the finitely many residual parents yield a single finite cutoff above which every realization in that exit layer descends. Combining this finite-layer control with residual mass collapse shows that the set of initially residual trajectories that never descend has natural density zero.

The remaining congruence classes require no residual analysis. Positive even integers descend immediately under the ordinary Collatz map, while odd integers congruent to 1 (mod 4), apart from the trivial value 1, descend after one fully accelerated step. The only nontrivial initial class is

$$x \equiv 3 \pmod{4},$$

which is precisely the realization set $\mathcal{R}_1$ of the root word (1) of the residual forest. Accelerated descent for an odd integer occurs at an actual iterate of the ordinary Collatz map, so the accelerated result transfers directly to ordinary eventual descent.

Hence

$$\lim_{N \rightarrow \infty} \frac{\#\{x \in \mathbb{N}_{>0} : x < N, \exists m \geq 0, T^m(x) < x\}}{N} = 1.$$

The proof developed here is organized through a sequence of exact mathematical structures:

$$\begin{aligned} \text{division words} &\longrightarrow \text{affine realization} \longrightarrow \text{dyadic cylinders} \\ &\longrightarrow \text{residual forest} \longrightarrow \text{residual mass collapse} \longrightarrow \text{eventual descent.} \end{aligned}$$

Several ingredients underlying this framework—including affine representations of finite Collatz trajectories, finite residue coding, and density-one finite stopping—have classical antecedents in the work of Terras and Everett. The formulation used here reorganizes these ideas in terms of exact accelerated division counts and a residual-cylinder decomposition, leading to the residual-mass argument above. At each stage the symbolic description is tied to exact arithmetic rather than to a heuristic trajectory model, and the final density argument uses only finite-stage decompositions and ordinary natural-density limits.

The result does not prove that every Collatz trajectory descends below its starting value, nor does it prove convergence of every trajectory to 1. A density-zero exceptional set may remain nonempty or infinite. What is proved is that eventual descent below the initial value holds for a set of positive integers of natural density one.

A Explicit Computation of the Affine Map

Each division word

$$\omega = (d_0, \dots, d_{m-1})$$

determines the formal affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}, \quad D(\omega) = \sum_{i=0}^{m-1} d_i,$$

where the affine constant $c(\omega)$ is determined uniquely by symbolic composition.

By Theorem 4.4, the constant is given explicitly by

$$c(\omega) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{\sum_{i=0}^{j-1} d_i}.$$

This is an integer identity, not a choice of residue representative modulo $2^{D(\omega)}$. In particular, $c(\omega)$ need not lie in the interval

$$0 \leq c(\omega) < 2^{D(\omega)}.$$

For example, if

$$\omega = (1, 1),$$

then

$$c(\omega) = 3 + 2 = 5,$$

and therefore

$$F_\omega(x) = \frac{9x + 5}{4}.$$

Although

$$5 \equiv 1 \pmod{4},$$

replacing 5 by 1 would change the affine map:

$$\frac{9x + 1}{4} \neq \frac{9x + 5}{4}.$$

Thus the affine constant must be retained as the actual integer produced by symbolic composition rather than reduced modulo the denominator.

A.1 A Four-Step Example

Consider

$$\omega = (1, 1, 1, 4).$$

Then

$$m = 4, \quad D(\omega) = 7,$$

and

$$\begin{aligned} c(\omega) &= 3^3 + 3^2 2^1 + 3 \cdot 2^{1+1} + 2^{1+1+1} \\ &= 27 + 18 + 12 + 8 \\ &= 65. \end{aligned}$$

Hence

$$\boxed{F_\omega(x) = \frac{81x + 65}{128}.}$$

The formal-integrality condition is

$$81x + 65 \equiv 0 \pmod{128},$$

which gives

$$x \equiv 15 \pmod{128}.$$

Thus every integer of the form

$$x = 15 + 128k$$

makes the formal affine expression integral, and

$$F_\omega(15 + 128k) = 81k + 10.$$

Formal integrality, however, is weaker than exact realization. By Theorem 5.5, exact realization is equivalent to

$$81x + 65 \equiv 128 \pmod{256},$$

which says precisely that the quotient remaining after removal of the prescribed total power 2^7 is odd. Solving gives

$$\boxed{x \equiv 143 \pmod{256}.}$$

Thus the formal-integrality class

$$15 \pmod{128}$$

splits into two classes modulo 256. Only

$$143 \pmod{256}$$

realizes the division word $(1, 1, 1, 4)$.

This distinction can also be seen directly from

$$F_\omega(15 + 128k) = 81k + 10.$$

The endpoint is odd exactly when k is odd. Writing

$$k = 2j + 1$$

gives

$$x = 15 + 128(2j + 1) = 143 + 256j,$$

which is precisely the exact realization class.

For the canonical representative $x = 143$,

$$F_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91.$$

The accelerated trajectory is

$$143 \mapsto 215 \mapsto 323 \mapsto 485 \mapsto 91,$$

with successive division counts

$$(1, 1, 1, 4).$$

A.2 Direct Derivation by Symbolic Substitution

For completeness, consider a general four-entry division word

$$\omega = (a, b, c, d).$$

The formal symbolic relations are

$$x_1 = \frac{3x_0 + 1}{2^a}, \quad x_2 = \frac{3x_1 + 1}{2^b}, \quad x_3 = \frac{3x_2 + 1}{2^c}, \quad x_4 = \frac{3x_3 + 1}{2^d}.$$

Eliminating the intermediate variables gives

$$2^{a+b+c+d}x_4 = 3^4x_0 + 3^3 + 3^22^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

Therefore

$$F_\omega(x) = \frac{3^4x + 3^3 + 3^22^a + 3 \cdot 2^{a+b} + 2^{a+b+c}}{2^{a+b+c+d}},$$

and hence

$$\boxed{c(\omega) = 3^3 + 3^22^a + 3 \cdot 2^{a+b} + 2^{a+b+c}.$$

This calculation illustrates how each successive “+1” term is propagated through the later affine steps. Earlier contributions acquire additional factors of 3, while the powers of 2 record the prescribed divisions occurring before each contribution reaches the final numerator.

The example also illustrates the distinction used throughout the paper: the affine constant is determined by symbolic composition, formal integrality is a congruence modulo $2^{D(\omega)}$, and, for admissible words, exact realization is characterized by the strengthened congruence modulo $2^{D(\omega)+1}$.

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